

Boundary dynamics, triple transitivity, and mixed identities in weakly hyperbolic groups

Ekaterina Rybak

Abstract

We study the interplay between the algebraic and dynamical properties of groups that admit a general type action on a δ -hyperbolic space such that the induced action on the limit set in the Gromov boundary is faithful. We divide the class of such groups into two subclasses based on a dynamical criterion: groups whose induced action on the limit set is topologically free, and those whose action is not. We prove that satisfying the criterion is equivalent to a purely algebraic property of being mixed identity free, generalizing results of P. Fima, F. Le Maître, S. Moon, and Y. Stalder from groups acting on trees to groups acting on arbitrary hyperbolic spaces. As one corollary, we obtain that the reduced C^* -algebra of such groups is selfless, using the results of N. Ozawa. For the subclass of groups whose action on the limit set is not topologically free, we prove the rigidity result for all 3-transitive faithful actions and bound the transitivity degree by 3, generalizing the result of A. Le Boudec and N. Matte Bon. As an application, we show that non-affine simple Kac-Moody groups over finite fields are MIF, answering the question of J. Belk, F. Fournier-Facio, J. Hyde, and M. Zaremsky.

Contents

1	Introduction	2
1.1	Weakly hyperbolic groups	2
1.2	MIF weakly hyperbolic groups	4
1.3	Rigidity of 3-transitive actions	7
2	Preliminary results	8
2.1	Group actions on hyperbolic spaces	8
2.2	Minimal actions on trees	15
2.3	Elliptic radical	16
2.4	Micro-supported actions	18
3	Boundary dynamics and algebraic properties	21
3.1	The cross-ratio and topological transitivity of actions on the limit set	21
3.2	Boundary dynamics and mixed identities	24

MSC Primary: 20F65, 20F69. Secondary: 20F67, 46L05

3.3	Boundary dynamics and triple transitivity	29
3.4	Examples and necessity of assumptions	38

1 Introduction

1.1 Weakly hyperbolic groups

One of the generalizations of the class of non-elementary hyperbolic groups is the class of acylindrically hyperbolic groups. Recently, studies of acylindrically hyperbolic groups have attracted a lot of attention. A group is *acylindrically hyperbolic* if it admits an acylindrical action of general type on a hyperbolic space. The acylindricity condition is a generalization of properness of the action. We refer the reader to [Osi16] for the definitions. The class of acylindrically hyperbolic groups is large; in particular, it includes all non-elementary hyperbolic and relatively hyperbolic groups, mapping class groups of punctured closed surfaces except for surfaces with genus $g = 0$ and fewer than three punctures, $Out(F_n)$ for $n \geq 2$, directly indecomposable right-angled Artin groups, and 1-relator groups with at least 3 generators. Despite being such a broad class, the requirement of the acylindricity of the action is restrictive enough to obtain nontrivial results about the algebraic properties of acylindrically hyperbolic groups. For example, acylindrically hyperbolic groups are closed under taking infinite normal subgroups [Osi16], are SQ-universal, and have finite amenable radical [DGO]. Finitely generated acylindrically hyperbolic groups have cut points in all of their asymptotic cones [Sis] and have exponential conjugacy growth [HO13].

We want to enlarge the class of acylindrically hyperbolic groups even further and study what properties hold when the acylindricity condition is removed. Following the terminology from [MT], we say

Definition 1.1. A group G is *weakly hyperbolic* if it admits a general type action on a geodesic hyperbolic space. An isometric action on a hyperbolic space is *of general type* if orbits are unbounded and G contains two loxodromic elements with disjoint fixed points on the Gromov boundary.

We refer the reader to Section 2.1 for details on Gromov hyperbolic spaces and the classification of group actions on them. Examples of weakly hyperbolic groups that are not necessarily acylindrically hyperbolic come from groups acting on trees and, more generally, groups of isometries of hyperbolic spaces.

Many properties of particular examples of weakly hyperbolic groups are well documented in the literature. Various results on random walks in weakly hyperbolic groups were obtained in [MT], [Sun], [BMSS], [AMS], [AS], [GTT22], [Cho], and [Hor]. However, not many group structural results are known about the whole class. Using a simple ping-pong argument, one can prove that any weakly hyperbolic group contains a nonabelian free subgroup (see [Gro87, 8.2.F] or [Ham, Thm. 2.7]). In [Osi17], it is proved that for every invariant random subgroup H of a countable weakly hyperbolic group G , when G acts on some hyperbolic

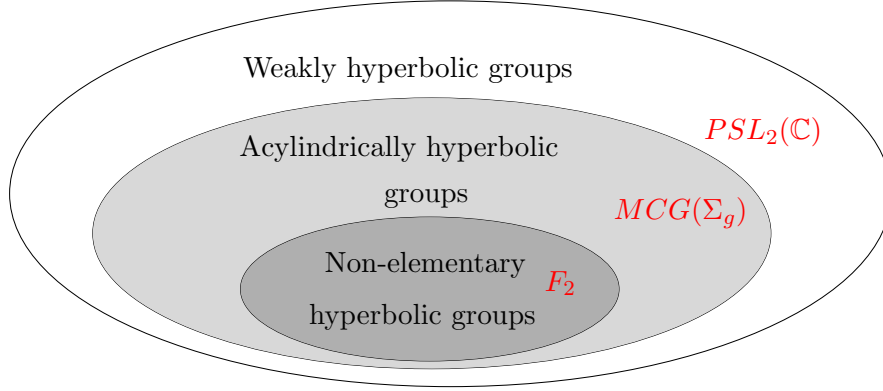


Figure 1: The class of weakly hyperbolic groups. By F_2 we denote the free group on 2 generators, $MCG(\Sigma_g)$ denotes the mapping class group of any punctured closed surface, except for surfaces with genus $g = 0$ and fewer than three punctures, $PSL_2(\mathbb{C})$ denotes the projective special linear group over $\mathbb{C}$.

space via a general type action, the induced action of H either has the same limit set as G or all orbits of H are bounded. In particular, every normal subgroup of a weakly hyperbolic group satisfies this dichotomy. In [GOR, Thm 1.2] it is proved that the Frattini subgroup of a countable weakly hyperbolic group acts on a hyperbolic space with bounded orbits, and thus has infinite index.

A natural question to ask is what other interesting algebraic properties are satisfied by every weakly hyperbolic group. Finding such properties is challenging, as having just the general type action without any properness condition is not very limiting. Therefore, we want to impose an additional condition on the group action.

Recall that the limit set $\Lambda_S(G)$ in the Gromov boundary of a hyperbolic space S is the closure of the set of accumulation points of orbits under the G -action.

Definition 1.2. We say that a group G is *faithful weakly hyperbolic* if it admits a general type action on a hyperbolic space S such that the induced G -action on $\Lambda_S(G)$ is faithful.

Note that not every weakly hyperbolic group is faithful weakly hyperbolic. A source of examples of weakly hyperbolic groups that are not faithful weakly hyperbolic groups is given in Proposition 3.31. However, by Proposition A, any weakly hyperbolic group has a faithful weakly hyperbolic quotient. Here, $E_S(G)$ is the *elliptic radical* with respect to the action of G on S . It is the set of all elements of G that act on $\Lambda_S(G)$ trivially. We refer the reader to Section 2.3 for the basic properties of $E_S(G)$.

Proposition A. Suppose that a group G admits a general type action on a hyperbolic space S . Then $G/E_S(G)$ is faithful weakly hyperbolic.

Let X be a topological space. Recall that an action of a group G on X by homeomorphisms is *topologically free* if the set of fixed points $\text{Fix}_X(g)$ of every non-trivial element

$g \in G$ has empty interior.

Definition 1.3. A weakly hyperbolic group G is called *lim-free* if it admits a general type action on a hyperbolic space S such that the induced action on $\Lambda_S(G)$ is topologically free.

We have an obvious inclusion: Lim-free weakly hyperbolic groups are faithful weakly hyperbolic groups. While the property of being weakly hyperbolic is not closed under taking non-trivial normal subgroups, both properties of being faithful weakly hyperbolic and of being lim-free are closed under taking non-trivial normal subgroups by Lemma 2.26. We will see that lim-free weakly hyperbolic groups and faithful weakly hyperbolic groups that are not lim-free are two distinct classes. Groups from the former class include acylindrically hyperbolic groups without non-trivial finite normal subgroups (see Corollary 3.12) and behave similarly to them, while groups from the latter class have very different algebraic properties.

1.2 MIF weakly hyperbolic groups

Let F_n denote a free group of rank n . Recall that a group G satisfies a *mixed identity* $w = 1$ for some $w \in G * F_n$ if every homomorphism $G * F_n \rightarrow G$ that is the identity on G sends w to 1. A mixed identity $w = 1$ is *non-trivial* if w is a non-trivial element of $G * F_n$, and a group G is *mixed identity free* (or MIF for brevity) if it does not satisfy any non-trivial mixed identity. The property of being MIF imposes strong restrictions on the algebraic structure of G . If G is MIF, then G has infinite conjugacy classes, it is directly indecomposable, and it has infinite girth (see [HO16, Prop. 5.4]). A countable group G is MIF if and only if G and $G * F_n$ are universally equivalent as G -groups for all $n \in \mathbb{N}$ (see [HO16, Prop. 5.3]). A surprising connection between MIF groups and the Boone-Higman conjecture was recently discovered in [BFFHZ].

In our first main theorem, part (b) reveals a rigidity phenomenon: properties of one general type action impose strong constraints on all general type actions. Furthermore, part (c) shows that the geometrically defined class of lim-free groups can be characterized within the class of faithful weakly hyperbolic groups by a purely algebraic condition.

Theorem B. *For any faithful weakly hyperbolic group G , the following conditions are equivalent:*

- (a) G is *lim-free*;
- (b) for any general type action of G on a hyperbolic space with trivial elliptic radical, the induced action on the limit set is topologically free;
- (c) G is MIF.

Considering mixed identities in weakly hyperbolic groups was inspired by the results about groups acting on simplicial trees. An action of a group G on a tree T is called

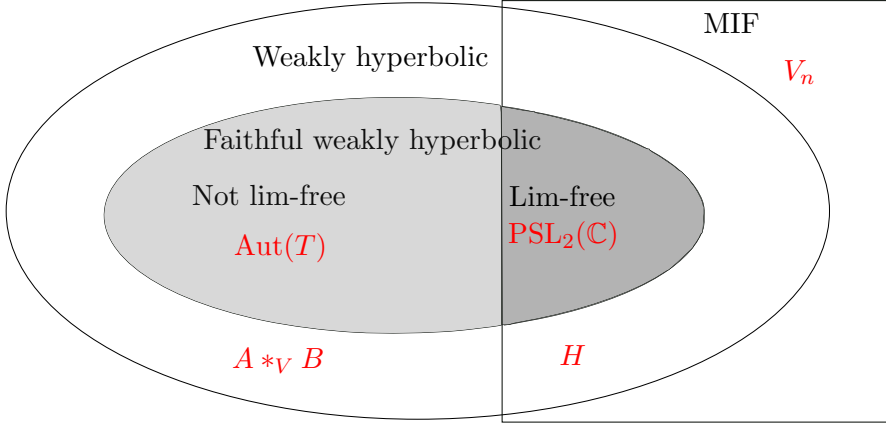


Figure 2: The interaction between classes of weakly hyperbolic groups and MIF groups. Here $A *_V B$ denotes the verbal product of non-trivial groups A and B with respect to a nonempty set of words V , where at least one of the factors is weakly hyperbolic (see Example 3.29), V_n denotes the Higman-Thompson group (see [BEH, Thm. 3.5]), and H stands for a group constructed in Example 3.40.

minimal if there are no nontrivial invariant subtrees. Combining [BM, Proposition 3.7] with their results, P. Fima, F. Le Maître, S. Moon, and Y. Stalder prove that if the action of a group G on a simplicial tree T is faithful, minimal, and of general type, then G is MIF if and only if the induced action on ∂T is topologically free [FMMS, Theorem B]. Their result can be obtained as a corollary of Theorem B (See Corollary 3.14).

One of the applications of Theorem B, not covered by [FMMS, Theorem B], is the proof that infinite simple Kac-Moody groups over finite fields are MIF (see Example 3.42). This partially answers Question 3.6 from [BFFHZ].

Let G be a group and $w(t) \in G * \langle t \rangle$ be a mixed identity. The solution set $\{g \in G \mid w(g) = 1\}$ is called *the primitive solution set corresponding to $w(t)$* . The Zariski topology on G is defined by taking the collection of primitive solution sets to be a subbasis for the closed sets of the topology. A Zariski-closed subgroup (or more generally, a subset) of G is called *algebraic*. The natural question to ask is what proper subgroups of a group G can be algebraic. If G is torsion-free non-elementary hyperbolic, then all such subgroups are cyclic [KM]. In [Jac], structural conditions for non-elementary algebraic subgroups of acylindrically hyperbolic groups were obtained. By a similar argument as in the proof of Theorem B, we get

Corollary C. *Let G be a lim-free weakly hyperbolic group. Then G does not contain any non-trivial proper normal Zariski-closed subgroups.*

Non-lim-free faithful weakly hyperbolic groups also have interesting common purely algebraic properties. For example, they contain infinite direct sums of infinite non-abelian subgroups (see Corollary 2.36). Recall that the *monolith* $\text{Mon}(G)$ of a group G is the

intersection of all non-trivial normal subgroups of G , and G is said to be *monolithic* if $\text{Mon}(G) \neq \{1\}$. As a corollary of [GR, Corollary 1.7], we obtain the following.

Proposition D. *Let G be a faithful weakly hyperbolic group. If G is not lim-free, then it is monolithic with an infinite simple monolith.*

In this way, non-lim-free faithful weakly hyperbolic groups are very different from acylindrically hyperbolic groups, whose monolith is always trivial (see Proposition 3.11). This allows us to obtain the result of M. Hull and D. Osin [HO16, Corollary 5.10] that acylindrically hyperbolic groups without nontrivial finite normal subgroups are MIF as a corollary of Theorem B and Proposition D (see Corollary 3.12).

Another corollary of Proposition D gives us a large class of lim-free weakly hyperbolic groups.

Corollary E. *Every residually finite faithful weakly hyperbolic group is lim-free. In particular, every finitely generated linear faithful weakly hyperbolic group is lim-free.*

The famous Tits Alternative states that a finitely generated linear group is either virtually solvable or contains a nonabelian free subgroup. As another corollary of Theorem B, we obtain an analogous statement for all subgroups of $\text{PSL}_2(\mathbb{C})$.

Corollary F. *Every subgroup of $\text{PSL}_2(\mathbb{C})$ is either virtually solvable or MIF.*

We should note that in [GGS], it is proved that any center-free unbounded and non-virtually solvable countable subgroup of $\text{SL}_2(k)$ is highly transitive, where k is a local field, which implies being MIF by the result of [HO16]. In [Tom], Tomanov gives necessary and sufficient conditions for Zariski-dense subgroups of linear groups to be MIF. Corollary F does not require any assumptions on subgroups of $\text{PSL}_2(\mathbb{C})$ and is not an immediate consequence of Corollary E, [Tom], or [GGS].

A property called *selflessness* was recently introduced for a C^* -algebra by L. Robert. It implies many important regularity properties such as simplicity, stable rank one in the tracial setting, and strict comparison. We refer the reader to [Rob] for the definitions. In [AGKEP], Tattwamasi Amrutam, David Gao, Srivatsav Kunnawalkam Elayavalli, and Gregory Patchell proved that acylindrically hyperbolic groups without finite normal subgroups have selfless reduced C^* -algebras. In their proof, they used a stronger quantitative version of the property of being MIF and the rapid decay property. This attracted attention to the quantitative version of being MIF, which resulted in a series of works: [BS], [AG], [Vid]. Later, in his remarkable paper, Ozawa obtained a dynamical condition for the group to have a selfless reduced C^* -algebra. Using the results of [Oza], we obtain

Corollary G. *If a weakly hyperbolic group G is lim-free, its reduced group C^* -algebra $C_\lambda^*(G)$ is selfless.*

Corollary G gives us the first examples of finitely presented simple groups with selfless reduced C^* -algebras.

Proposition H. *Non-affine simple finitely presented Kac-Moody groups over finite fields constructed in [CR] are MIF, and their reduced C^* -algebras are selfless.*

1.3 Rigidity of 3-transitive actions

Recall that a G -action on some set Ω is k -transitive for some $k \in \mathbb{N}$ if for any two k -tuples of distinct points $(\omega_1, \dots, \omega_k) \in \Omega^k$ and $(v_1, \dots, v_k) \in \Omega^k$, there exists $g \in G$ such that $g\omega_i = v_i$ for all $i = 1, \dots, k$. As discussed earlier, Theorem B gives a rigidity phenomenon for general type actions of lim-free weakly hyperbolic groups. If a group is faithful weakly hyperbolic but not lim-free, we obtain a rigidity phenomenon for all 3-transitive actions on any set, as our second main theorem shows.

Definition 1.4. Two actions of a group G on sets Ω and Ω' are *conjugate* if there exists a bijective G -equivariant map $\Omega \rightarrow \Omega'$.

Theorem I. *Suppose that the action of a group G on a hyperbolic space S is of general type and that the induced action of G on $\Lambda_S(G)$ is faithful and not topologically free. Assume that G acts faithfully and 3-transitively on a set Ω . Then there exists a G -orbit $\mathcal{O} \subset \Lambda_S(G)$ such that the actions of G on Ω and on $\mathcal{O}$ are conjugate.*

This theorem is the analogue of [BM, Theorem 1.4] for group actions on simplicial trees. We generalize their arguments to groups acting on arbitrary hyperbolic spaces and obtain their result as a corollary (see Corollary 3.28).

The *transitivity degree* of a group G is the maximum k such that G admits a k -transitive faithful action on some set. Among infinite groups, examples of groups with transitivity degree 1, 2, 3, and ∞ are known. Whether there exists an infinite group with transitivity degree 4 is an open question. The next corollary shows that if such an example exists, it cannot be a faithful weakly hyperbolic group that is not lim-free.

Corollary J. *Let G be a faithful weakly hyperbolic group that is not lim-free. Then the transitivity degree of G is at most 3.*

Recall that an action of a group G on a topological space X is *topologically transitive* if, for every pair of nonempty open subsets U and V of X , there is an element $g \in G$ such that $g(U) \cap V \neq \emptyset$. It is *topologically k -transitive* for $k \in \mathbb{N}$ if the induced action of G on the k -fold Cartesian product X^k is topologically transitive.

Since Theorem I tells us that the limit set “sees” all 3-transitive actions of a faithful weakly hyperbolic group G that is not lim-free, it suffices to examine just the G -action on the limit set to determine whether the transitivity degree of G is 3 or less, as the following corollary shows.

Corollary K. *Let G be a faithful weakly hyperbolic group that is not lim-free. If the transitivity degree of G is 3, then for any general type action of G on any hyperbolic space S , if the induced action on $\Lambda_S(G)$ is faithful, it is topologically 3-transitive.*

The paper is organized as follows. In Section 2, we provide the necessary background and preliminary results and prove Proposition A in Subsection 2.3. In Subsection 3.2, we prove Theorem B, Corollary C, Proposition D, Corollary E, Corollary F, and Corollary G. In Subsection 3.3, we prove Theorem I, Corollary J, and Corollary K. In Subsection 3.4, we prove Proposition H and discuss some examples and the necessity of the conditions of Theorem B and Theorem I.

Acknowledgments. The author is grateful to Denis Osin and Spencer Dowdall for providing guidance throughout the course of this project. The author is thankful to Francesco Fournier-Facio for comments on the first version, for drawing her attention to Kac–Moody groups, and for helpful discussions concerning Example 3.42.

2 Preliminary results

Notation. Throughout the paper, we denote by S a geodesic δ -hyperbolic metric space, and by d_S the metric on S . All group actions on S are by isometries. For a group G acting on S and any $s \in S$, let $Gs = \{gs \mid g \in G\}$ denote the G -orbit of the point s .

For group elements g and h we denote by g^h the conjugate $h^{-1}gh$. If a group G acts on a set X , we denote by $\text{Fix}_X(G)$ the set of fixed points of G in X , and by G_x the stabilizer subgroup of $x \in X$ under the action of G . If U is a subset of a topological space X , we denote by $\text{Int}(U)$ the interior of U in X .

2.1 Group actions on hyperbolic spaces

We refer the reader to [DSU, Chapter 3] and [BS07] for more details on the material discussed in this subsection.

Definition 2.1. Let S be a geodesic metric space. Recall that the *Gromov product* of two points $x, y \in S$ with respect to a point $o \in S$ is defined by

$$\langle x|y \rangle_o := \frac{1}{2}(d_S(x, o) + d_S(y, o) - d_S(x, y)).$$

Definition 2.2 (Gromov hyperbolicity). A geodesic metric space S is δ -hyperbolic if for any points $x, y, z, o \in S$, we have

$$\langle x|y \rangle_o \geq \min \{ \langle x|z \rangle_o, \langle z|y \rangle_o \} - \delta.$$

If $Y \subseteq S$ is a subspace, then for any constant $K \geq 0$, we denote the closed K -neighborhood of Y in S by $\mathcal{N}_K(Y) = \{x \in S \mid d_S(x, Y) \leq K\}$.

Definition 2.3 (Rips hyperbolicity). Fix $\delta \geq 0$. A geodesic metric space S is δ -hyperbolic if given any $x, y, z \in S$ and any geodesics α, β, γ between them, we have $\gamma \subseteq \mathcal{N}_\delta(\alpha \cup \beta)$.

A geodesic metric space is Rips hyperbolic if and only if it is Gromov hyperbolic (see, for example, [KS, Lemma 1.46]). In this work, we choose δ in a way that S satisfies both Definitions 2.2 and Definition 2.3 simultaneously.

A sequence (x_i) of elements of S *converges at infinity* if $\langle x_i | x_j \rangle_o \rightarrow \infty$ as $i, j \rightarrow \infty$ for some, or equivalently for any, point $o \in S$. Two such sequences (x_i) and (y_i) are *equivalent* if $\langle x_i | y_j \rangle_o \rightarrow \infty$ as $i, j \rightarrow \infty$. The *Gromov boundary* of S , denoted by ∂S , is defined as the set of equivalence classes of sequences converging at infinity. If a is the equivalence class of (x_i) , we say that the sequence (x_i) converges to a . There is a natural topology on $\widehat{S} = S \cup \partial S$ extending the topology on S such that S is dense in $\widehat{S}$, which will be discussed later. The definition of ∂S is independent of the choice of the basepoint, and any isometric group action on S extends to a continuous action on $\widehat{S}$. The Gromov product can be extended to $\widehat{S}$ in the following way.

Definition 2.4. Let $a, b \in \partial S$ and $o, x \in S$.

1. The *Gromov product* of a and b with respect to o is defined by

$$\langle a | b \rangle_o = \inf \left\{ \liminf_{i, j \rightarrow +\infty} \langle x_i | y_j \rangle_o \mid (x_i) \rightarrow a, (y_j) \rightarrow b \right\}.$$

2. The *Gromov product* of a and x with respect to o is defined by

$$\langle a | x \rangle_o = \inf \left\{ \liminf_{i \rightarrow \infty} \langle x_i | x \rangle_o \mid (x_i) \rightarrow a \right\}.$$

The next lemma shows that taking the $\liminf$ in the definition of the Gromov product of the boundary points is the same as taking $\limsup$ up to a constant and that the boundary points also satisfy the four-point condition.

Lemma 2.5 ([BS07], Lemma 2.2.2). *Let S be a δ -hyperbolic space. Let $o \in S$, and let $a, b, c \in \partial S$.*

1. *For arbitrary sequences $(x_i) \rightarrow a, (y_i) \rightarrow b$, we have*

$$\langle a | b \rangle_o \leq \liminf_{i \rightarrow \infty} \langle x_i | y_i \rangle_o \leq \limsup_{i \rightarrow \infty} \langle x_i | y_i \rangle_o \leq \langle a | b \rangle_o + 2\delta.$$

- 2.

$$\langle a | b \rangle_o \geq \min \{ \langle a | c \rangle_o, \langle c | b \rangle_o \} - \delta.$$

The Gromov product of distinct points of the boundary is always finite, i.e., if $a, b \in \partial S$, then $\langle a | b \rangle_o = \infty$ if and only if $a = b$. It is also preserved by isometries of S , i.e., if g is an isometry of S , $a, b \in \widehat{S}$ then $\langle ga | gb \rangle_{go} = \langle a | b \rangle_o$ (see [DSU, Section 3.4.1]).

For any point $a \in \partial S$, $o \in S$, and $r \in \mathbb{R}$, denote by

$$\mathcal{U}_o(a, r) = \left\{ s \in \widehat{S} \mid \langle a | s \rangle_o > r \right\}.$$

The subset $\mathcal{U}_o(a, r)$ is open in the topology on $\widehat{S}$, and $\{\mathcal{U}_o(a, r) \mid r \geq 0\}$ form a neighborhood base for the topology on $\widehat{S}$ at a (see [DSU, Remark 3.4.15 (II)]).

The *limit set* $\Lambda_S(G)$ of G is defined as the intersection of the closure of Gs in $\widehat{S}$ with ∂S . This definition is independent of the choice of $s \in S$.

An element $g \in G$ is said to be *loxodromic* if the group $\langle g \rangle$ has unbounded orbits and fixes exactly two points on ∂S (equivalently, $|\Lambda_S(\langle g \rangle)| = 2$). By $\mathcal{L}_S(G)$ we denote the set of all loxodromic elements of G .

The following lemma establishes the standard notation for fixed points of loxodromic elements, motivated by the so-called north-south dynamics (see [Gro87, Lemma 8.1.G] or [DSU, Theorem 6.1.10]).

Lemma 2.6. *The fixed points of $g \in \mathcal{L}_S(G)$ can be denoted by g^+ and g^- so that*

$$\lim_{n \rightarrow \infty} g^n s = g^+ \quad \forall s \in \widehat{S} \setminus \{g^-\} \quad \text{and} \quad \lim_{n \rightarrow \infty} g^{-n} s = g^- \quad \forall s \in \widehat{S} \setminus \{g^+\}$$

where the convergence in the first (respectively, second) limit is uniform on sets whose closure does not contain g^- (respectively, g^+).

Remark 2.7. If $g \in \mathcal{L}_S(G)$, then $fgf^{-1} \in \mathcal{L}_S(G)$ for any $f \in G$. Moreover, we have $(fgf^{-1})^+ = fg^+$ and $(fgf^{-1})^- = fg^-$.

We denote the set of endpoints of loxodromic elements by $\mathcal{H}_S(G) = \{g^\pm \mid g \in \mathcal{L}_S(G)\}$.

Possible actions of groups on hyperbolic spaces can be classified as follows according to the cardinality of $\Lambda_S(G)$ (see [Gro87, Sections 8.1-8.2], [Ham], or [DSU, Chapter 6]):

1. (*elliptic action*) $|\Lambda_S(G)| = 0$. Equivalently, G has bounded orbits.
2. (*parabolic action*) $|\Lambda_S(G)| = 1$. Equivalently, G has unbounded orbits, and $\mathcal{L}_S(G) = \emptyset$. In this case $\text{Fix}_{\partial S}(G) = \Lambda_S(G)$.
3. (*lineal action*) $|\Lambda_S(G)| = 2$. Equivalently, $\mathcal{L}_S(G) \neq \emptyset$, and any two loxodromic elements have the same fixed points on ∂S . In this case $\text{Fix}_{\partial S}(G) \subseteq \Lambda_S(G)$.
4. $|\Lambda_S(G)| = \infty$. Then $\mathcal{L}_S(G) \neq \emptyset$. In turn, this case breaks into two subcases.
 - (a) (*quasi-parabolic action*) $\text{Fix}_{\partial S}(G) \neq \emptyset$. Then G fixes a unique point of ∂S .
 - (b) (*general type action*) $\text{Fix}_{\partial S}(G) = \emptyset$. Equivalently, G contains at least two (or infinitely many) loxodromic elements with disjoint fixed points.

The action of G is called *elementary* in cases 1 – 3 and *non-elementary* in case 4.

Recall that an action of a group G by homeomorphisms on a topological space X is *minimal* if there is no proper closed nonempty G -invariant subset $Y \subset X$.

Corollary 2.8. [DSU, Corollary 7.4.1] *Suppose that the action of a group G on a hyperbolic space S is of general type. Then $\Lambda_S(G)$ is the smallest nonempty closed G -invariant subset of ∂S . Therefore, the action of G on $\Lambda_S(G)$ is minimal.*

The following can be found in [Gro87, Section 8.2]; alternatively, see [Ham, Theorems 2.6, 2.9] or [DSU, Propositions 7.4.6, 7.4.7].

Lemma 2.9. *Suppose that a group G acts non-elementarily on a hyperbolic space S . Then $\mathcal{H}_S(G)$ is dense in $\Lambda_S(G)$. Moreover, if the action is of general type, then $\{(g^+, g^-) \mid g \in \mathcal{L}_S(G)\}$ is dense in $\Lambda_S(G) \times \Lambda_S(G)$.*

The following lemma follows from [Osi17, Theorem 4.5], where a much more general fact is proved.

Lemma 2.10. *Suppose that a group G admits a general type action on a hyperbolic space S . Then the induced action of any normal subgroup $N \leq G$ either has bounded orbits in S or is also of general type. If the action of N on $\Lambda_S(G)$ is nontrivial, then $\Lambda_S(N) = \Lambda_S(G)$.*

Remark 2.11. The proof of Lemma 2.10 remains true for an arbitrary group G with a general type action on a hyperbolic space S , not necessarily countable as stated in [Osi17, Theorem 4.5].

For any $\lambda \geq 1$ and $c \in \mathbb{R}$, a map of metric spaces $\varphi : (X, d_X) \rightarrow (Y, d_Y)$ is a (λ, c) -quasi-isometric embedding if for all $x, y \in X$

$$\frac{1}{\lambda} d_X(x, y) - c \leq d_Y(\varphi(x), \varphi(y)) \leq \lambda d_X(x, y) + c.$$

A (λ, c) -quasi-geodesic is a (λ, c) -quasi-isometric embedding φ of an interval $I \subseteq \mathbb{R}$ into S . In this work, we will not distinguish between quasi-geodesics and their images when the parametrization is not important. If the constants λ and c of φ are not important, we write that φ is a quasi-geodesic.

Let $Y \subseteq S$, and let $\sigma \geq 0$. We say that Y is σ -quasiconvex in S if every geodesic with endpoints in Y is contained in the σ -neighborhood of Y . A subset $Y \subset S$ is said to be quasiconvex if it is σ -quasiconvex for some $\sigma \geq 0$. Any (λ, c) -quasi-geodesic in S is quasiconvex by the Morse Lemma.

Lemma 2.12 (Morse Lemma). *Let S be a δ -hyperbolic metric space, and fix $\lambda \geq 1$ and $c \geq 0$. There exists a constant σ depending only on δ, λ , and c such that if γ_1 and γ_2 are (λ, c) -quasi-geodesics in S with the same endpoints, then $\gamma_1 \subseteq \mathcal{N}_\sigma(\gamma_2)$.*

Let $Y \subset S$, and let $x \in S$. The nearest-point projection of x onto Y is a point $\bar{x}$ such that $d_S(x, \bar{x}) \leq d_S(x, Y) + 1$. When Y is σ -quasiconvex, the nearest-point projection is coarsely well-defined: for different choices $\bar{x}_1, \bar{x}_2$ of points in Y nearest to x we have $d_S(\bar{x}_1, \bar{x}_2) \leq \max(2, 2\sigma + 9\delta)$ (see [KS, Remark 1.103]).

Let x and y have nearest-point projections $\bar{x}$ and $\bar{y}$ onto a quasi-geodesic segment γ . The following lemma shows that if $\bar{x}$ and $\bar{y}$ are $O(\delta)$ apart, then, up to additive error, the geodesic from x to y goes from x to $\bar{x}$, then runs along the geodesic from $\bar{x}$ to $\bar{y}$, and then heads back out to y .

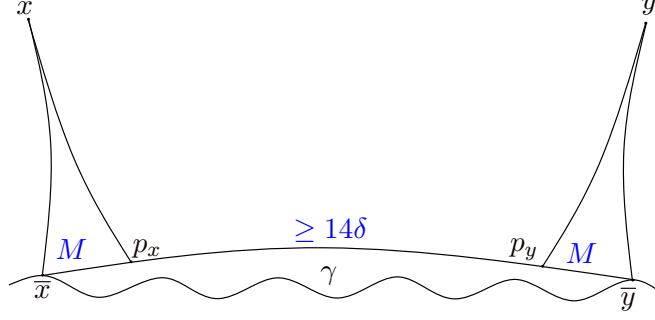


Figure 3: For Lemma 2.13

Lemma 2.13. *Let $x, y \in S$ and let $\gamma \subset S$ be a (λ, c) -quasi-geodesic segment. Let $\bar{x}$ be a nearest-point projection of x on γ and let $\bar{y}$ be a nearest-point projection of y on γ . Then there exists a constant $M = M(\lambda, c, \delta)$ such that if $d_S(\bar{x}, \bar{y}) > 14\delta + 2M$ then*

$$d_S(x, y) \geq d_S(x, \bar{x}) + d_S(\bar{x}, \bar{y}) + d_S(\bar{y}, y) - 24\delta - 4M.$$

Proof. Consider the geodesic segment $[\bar{x}, \bar{y}]$, and let p_x and p_y be nearest-point projections to $[\bar{x}, \bar{y}]$ of x and y , respectively. By [Mah, Proposition 3.6], if γ lies in a bounded neighborhood of a geodesic, then the nearest-point projections of any point to γ and to the geodesic are a bounded distance apart. Combining this with the Morse Lemma, we obtain that p_x, p_y lie within $M = M(\lambda, c, \delta)$ of $\bar{x}$ and $\bar{y}$. Since $d_S(\bar{x}, \bar{y}) > 14\delta + 2M$, we have $d_S(p_x, p_y) > 14\delta$. Therefore, by [Mah, Proposition 3.4], the following holds

$$d_S(x, y) \geq d_S(x, p_x) + d_S(p_x, p_y) + d_S(p_y, y) - 24\delta.$$

Using that $d_S(\bar{x}, p_x) \leq M$ and $d_S(\bar{y}, p_y) \leq M$, we obtain the conclusion of the lemma. $\square$

Definition 2.14. Let $g \in G$. The *asymptotic translation length* of g in S is

$$\tau_S(g) := \lim_{n \rightarrow \infty} \frac{d_S(s_0, g^n s_0)}{n}$$

for some (equivalently, any) $s_0 \in S$.

An element $g \in \mathcal{L}_S(G)$ if and only if $\tau_S(g) > 0$ [CDP, Chapitre 10, Proposition 6.3], and it acts on S by translation along a quasi-geodesic axis which connects the two limit points g^+ and g^- of g in ∂S .

If $\tau_S(g)$ is sufficiently large, such an axis can be chosen uniformly, as shown in the following lemma, which summarizes results from [Cou16, Section 3].

Lemma 2.15. *Let G act on a δ -hyperbolic space S . Suppose $g \in \mathcal{L}_S(G)$ and $\tau_S(g) \geq L_S \delta - 16\delta$, where L_S depends only on δ (and is more explicitly described in [Cou16, Definition 2.8]). Then there exists a $(2, \delta)$ -quasi-geodesic L_g in S which connects the limit points g^+ and g^- of g in ∂S . L_g is preserved by g and is $(\delta + 8\delta)$ -quasiconvex. The quasi-geodesic L_g is rectifiable, i.e., the length $l(\gamma)$ is well-defined for any segment $\gamma \subset L_g$.*

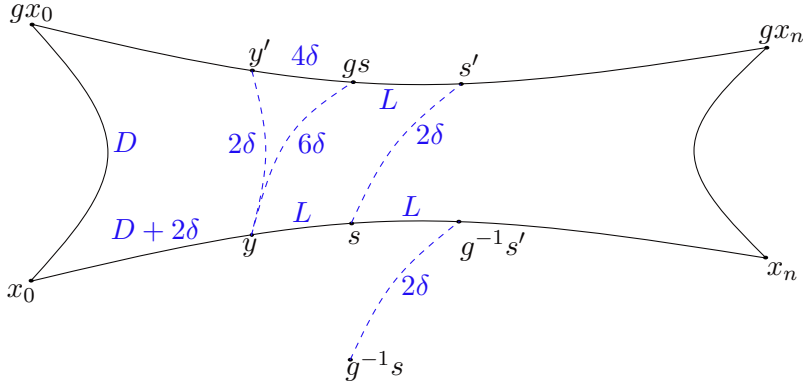


Figure 4: For Lemma 2.18

We will call such a uniform quasi-geodesic axis a *standard quasi-geodesic axis* of g . Note that we can always increase the asymptotic translation length by taking the appropriate power of g . Thus, the standard quasi-geodesic axis exists for every loxodromic element in G , up to passing to powers.

The following lemma is a reformulation of the “local-to-global” property in terms of Gromov products and the reverse triangle inequality for the orbits of loxodromic elements.

Lemma 2.16. *For every $A \in \mathbb{N}$ and every isometry $g \in G$ of S , if*

$$\langle gs \mid g^{-1}s \rangle_s \leq A \text{ and } d_S(s, gs) > 4A + 24\delta,$$

for some $s \in S$, then $g \in \mathcal{L}_S(G)$.

Proof. Consider the sequence $s, gs, g^2s, \dots, g^ns$ for some $n \in \mathbb{N}$. By [ABDHMW, Lemma 2.1], we have

$$d_S(s, g^ns) \geq \frac{1}{2} \sum_{j=1}^n d_S(g^{j-1}s, g^js). \quad (1)$$

Letting $n \rightarrow \infty$ in (1), and using that $d_S(s, gs) > 4A + 24\delta$, we conclude that $\tau_S(g) \geq 2A + 12\delta$. Since $\tau_S(g) > 0$, $g \in \mathcal{L}_S(G)$. $\square$

Proposition 2.17. *Suppose that the action of a group G on a hyperbolic space S is parabolic. Then for every $g_1, g_2 \in G$, there exists $s \in S$ such that $d_S(g_1s, s) \leq 56\delta$ and $d_S(g_2s, s) \leq 56\delta$.*

For the proof, we will need the following technical lemma.

Lemma 2.18. *Let G be a group acting on S by isometries. Let $(x_i)_{i \in \mathbb{N}}$ be a sequence in S converging to a point in ∂S . Fix a basepoint x_0 so that $\langle x_i \mid x_j \rangle_{x_0} \rightarrow \infty$ as $i, j \rightarrow \infty$.*

and an arbitrary $g \in G$. Let $D = d_S(x_0, gx_0)$. For a given $n \in \mathbb{N}$ consider the geodesic quadrilateral x_0, x_n, gx_n, gx_0 . For $s \in [x_0, x_n]$, if

$$2D + 4\delta < d_S(x_0, s) < \langle x_n \mid gx_n \rangle_{x_0} - 2D - 4\delta, \quad (2)$$

then $\langle g^{-1}s \mid gs \rangle_s \leq 8\delta$.

Proof. First, observe that for $s \in [x_0, x_n]$, if

$$D + 2\delta < d_S(x_0, s) < \langle x_n \mid gx_n \rangle_{x_0} - 2\delta, \quad (3)$$

then s is within 2δ of some point $s' \in [gx_0, gx_n]$. Indeed, any geodesic quadrilateral in S is 2δ -slim, so s is within 2δ of a point s' on one of the sides $[x_0, gx_0]$, $[gx_0, gx_n]$, or $[gx_n, x_n]$. If $s' \in [x_0, gx_0]$, then

$$d_S(x_0, s) \leq d_S(x_0, s') + d_S(s', s) \leq D + 2\delta,$$

which contradicts the first inequality in (3). If $s' \in [x_n, gx_n]$, then

$$\langle x_n \mid gx_n \rangle_{x_0} \leq d_S(x_0, s') \leq d_S(x_0, s) + 2\delta,$$

which contradicts the second inequality in (3).

Therefore, if (2) is satisfied, then there exists $s' \in [gx_0, gx_n]$ such that $d_S(s, s') \leq 2\delta$. Consider the point $gs \in [gx_0, gx_n]$. Note that, since $d_S(x_0, gx_0) = D$ and $d_S(s, s') \leq 2\delta$, by the triangle inequality, we obtain

$$|d_S(gx_0, s') - d_S(x_0, s)| \leq D + 2\delta.$$

Thus, we have

$$|d_S(gx_0, s') - d_S(gx_0, gs)| \leq D + 2\delta.$$

Since gx_0, gs, s' are on the same geodesic $[gx_0, gx_n]$, we conclude that $L := d_S(gs, s') \leq D + 2\delta$. Consider the point $g^{-1}s' \in [x_0, x_n]$. We have $d_S(s, g^{-1}s') = L$. Let y be the other point on the geodesic $[x_0, x_n]$ at distance L from s . Since $L \leq D + 2\delta$, by (2), we obtain that

$$D + 2\delta < d_S(x_0, y) < \langle x_n \mid gx_n \rangle_{x_0} - 2\delta.$$

Therefore, y is within 2δ of some point $y' \in [gx_0, gx_n]$. Note that since $d_S(s, s') \leq 2\delta$, by the triangle inequality $|d_S(y', s') - d_S(y, s)| \leq 4\delta$. Thus, we obtain

$$L - 4\delta \leq d_S(y', s') \leq L + 4\delta.$$

Since $d_S(gs, s') = L$, and y', gs, s' are all on the same geodesic, we get $d_S(y', gs) \leq 4\delta$. We have $d_S(g^{-1}s, g^{-1}s') = d_S(s, s') \leq 2\delta$, and $d_S(gs, y) \leq d_S(gs, y') + d_S(y, y') \leq 6\delta$. So,

$$\begin{aligned} \langle g^{-1}s, gs \rangle_s &= \frac{1}{2} (d_S(s, g^{-1}s) + d_S(s, gs) - d_S(g^{-1}s, gs)) \leq \\ &\leq \frac{1}{2} (d_S(s, g^{-1}s') + 2\delta + d_S(s, y) + 6\delta - d_S(g^{-1}s', y) + 8\delta) \leq \\ &\leq \langle y \mid g^{-1}s' \rangle_s + 8\delta = 8\delta. \end{aligned}$$

□

Proof of Proposition 2.17. Let q be the unique fixed point of G on ∂S and let $(x_i)_{i \in \mathbb{N}} \in S$ be a sequence converging to q . Fix a basepoint x_0 so that $\langle x_i | x_j \rangle_{x_0} \rightarrow \infty$ as $i, j \rightarrow \infty$. Now let arbitrary $g_1, g_2 \in G$ be given. Let $D = \max\{d_S(x_0, g_1 x_0), d_S(x_0, g_2 x_0)\}$, and take $n \in \mathbb{N}$ such that the geodesic segment $[x_0, x_n]$ is sufficiently long to satisfy the condition (2) in Lemma 2.18. By Lemma 2.18, we can find $s \in [x_0, x_n]$, such that

$$\langle g_1^{-1}s | g_1 s \rangle_s \leq 8\delta \text{ and } \langle g_2^{-1}s | g_2 s \rangle_s \leq 8\delta.$$

Since the action of G on S is parabolic, $\mathcal{L}_S(G) = \emptyset$. Therefore, by Lemma 2.16,

$$d_S(s, g_1 s) \leq 56\delta \text{ and } d_S(s, g_2 s) \leq 56\delta. \quad \square$$

Lemma 2.19. *Suppose that a group G acts by isometries on a hyperbolic space S . Let $g \in \mathcal{L}_S(G)$. Let L_g be the standard $(2, \delta)$ -quasi-geodesic axis of g . Let T be the translation distance of g along L_g . Then there exist $D = D(\delta) \in \mathbb{N}$ and $N \in \mathbb{N}$ such that for every $n \geq N$ and for every $s \in S$, we have*

$$2d_S(s, L_g) + \frac{n}{2}T - D \leq d_S(s, g^n s) \leq 2d_S(s, L_g) + 2nT + D.$$

Proof. First, choose $N \in \mathbb{N}$ such that for any $x \in L_g$ and $n \geq N$ the distance $d_S(x, g^n x) \geq 2M(\delta) + 14\delta$, where M is a constant from Lemma 2.13, applied to L_g . Take any $s \in S$. Let $\bar{s}$ be any nearest point projection of s on L_g . Then $g^n \bar{s} \in L_g$ and $g^n \bar{s}$ is the nearest-point projection of $g^n s$ onto L_g .

$$\begin{aligned} d_S(s, g^n s) &\geq d_S(s, \bar{s}) + d_S(\bar{s}, g^n \bar{s}) + d_S(g^n \bar{s}, g^n s) - 24\delta - 4M(\delta) \\ &\geq 2d_S(s, L_g) + d_S(\bar{s}, g^n \bar{s}) - 24\delta - 4M(\delta) - 2. \end{aligned} \tag{4}$$

Let γ be the segment of L_g from $\bar{s}$ to $g^n \bar{s}$. Since L_g is a $(2, \delta)$ -quasi-geodesic and T is the translation distance of g along L_g , we have

$$\frac{n}{2}T - \delta = \frac{1}{2}l(\gamma) - \delta \leq d_S(\bar{s}, g^n \bar{s}) \leq 2l(\gamma) + \delta = 2nT + \delta. \tag{5}$$

Combining (4), (5), and the triangle inequality, we obtain

$$2d_S(s, L_g) + \frac{n}{2}T - D \leq d_S(s, g^n s) \leq 2d_S(s, L_g) + 2nT + D,$$

where $D = 25\delta + 4M(\delta) + 2$. $\square$

2.2 Minimal actions on trees

We will use the facts established in this subsection to obtain the results about groups acting on simplicial trees from [BM] and [FMMS] as corollaries of Theorem B and Theorem I.

Proposition 2.20. [Bas, Proposition 7.5] Suppose that a group G acts on a simplicial tree T by isometries. If G contains a loxodromic element, then there is a unique minimal nonempty G -invariant subtree $L \subseteq T$, which is the union of the axes of all loxodromic elements.

The next lemma shows that if a group G , acting on a tree T , contains loxodromic elements, then the minimality of the action is almost equivalent to $\Lambda_T(G) = \partial T$, in the sense that every nontrivial invariant subtree is “large”.

Lemma 2.21. Suppose that the group G acts on a simplicial tree T by isometries, and $\Lambda_T(G) \neq \emptyset$. If the action of G on T is minimal, then $\Lambda_T(G) = \partial T$. Moreover, without the minimality assumption, the following conditions are equivalent:

1. $\Lambda_T(G) = \partial T$;
2. For every G -invariant subtree $S \subseteq T$, we have $\partial T = \partial S$.

Proof. By Proposition 2.20, the union of the axes of all loxodromic elements $L \subseteq T$ is the unique nonempty G -invariant subtree. Since the action of G on T is minimal, $L = T$. By the construction of L , the set of fixed points of loxodromic elements of G is dense in ∂L . Therefore, $\Lambda_L(G) = \partial L = \partial T$.

Now, assume that $\Lambda_T(G) = \partial T$. Let $S \subseteq T$ be a G -invariant subtree. Note that $\Lambda_T(G) \subseteq \partial S$ since $\Lambda_T(G)$ does not depend on a chosen point $t \in T$ and we can choose $t \in S$. Then we have $\partial T \subseteq \partial S$, so $\partial T = \partial S$. Conversely, assume that for every G -invariant subtree $S \subseteq T$, we have $\partial S = \partial T$. Let L be the union of the axes of all loxodromic elements. By Proposition 2.20, L is a G -invariant subtree of T , and $\Lambda_L(G) = \partial L$ by the construction of L . Since the limit set does not depend on a chosen point $t \in T$, we have $\Lambda_T(G) = \Lambda_L(G) = \partial L = \partial T$. $\square$

Lemma 2.22. Every elliptic subgroup of a group G acting on any tree T by isometries either fixes a vertex or a midpoint of an edge.

Proof. By [HV, Chapter 3.b, Proposition 9], every elliptic subgroup of G fixes some point of T , and, by [Bas, Proposition 7.2], this point should be either a vertex or a midpoint of an edge. $\square$

2.3 Elliptic radical

Definition 2.23. [Osi17] Given an action of a group G by isometries on a hyperbolic space S , define the elliptic radical of G with respect to this action by

$$E_S(G) = \{g \in G \mid gs = s \ \forall s \in \Lambda_S(G)\}.$$

Note that this definition makes sense for arbitrary actions on hyperbolic spaces. For general type actions, the following proposition provides an equivalent characterization.

Proposition 2.24. [Osi17, Proposition 3.4] *Suppose that a group G admits a general type action on a hyperbolic space S . Then $E_S(G)$ is the unique maximal elliptic normal subgroup of G .*

Recall that the *finite radical* $K(G)$ of an acylindrically hyperbolic group G is the unique maximal finite normal subgroup of G (see [DGO, Thm 6.14]).

Lemma 2.25. [BH, Lemma 3.7] *If the action of a group G on a hyperbolic space S is of general type and acylindrical, then $E_S(G) = K(G)$.*

The next lemma shows that the property of being faithful weakly hyperbolic and the property of being lim-free pass to normal subgroups; thus, all the results in this work pass to normal subgroups.

Lemma 2.26. *Let H be a non-trivial normal subgroup of a faithful weakly hyperbolic group G . Then H is also a faithful weakly hyperbolic group. If G is lim-free, then H is also lim-free.*

Proof. Suppose that the action of G on a hyperbolic space S is of general type and the induced action of G on $\Lambda_S(G)$ is faithful. The subgroup H is normal in G , so by Lemma 2.10, either the action of H on S is elliptic or it is of general type; in the latter case, $\Lambda_S(H) = \Lambda_S(G)$. Assume that H is elliptic. Since H is normal, $H \leq E_S(G)$ by the maximality of $E_S(G)$, and, thus, is trivial because the action of G on $\Lambda_S(G)$ is faithful, which is a contradiction. Therefore, the action of H on S is of general type, and the induced action of H on $\Lambda_S(H) = \Lambda_S(G)$ is faithful. If the action of G on $\Lambda_S(G)$ is topologically free, so is the action of H on $\Lambda_S(H)$. $\square$

We finish this subsection by proving Proposition A, which allows us to apply the main results of this work to some quotients of weakly hyperbolic groups that are not faithful weakly hyperbolic.

Proof of Proposition A. We need to show that there exists a hyperbolic space S' such that the action of $G/E_S(G)$ on S' is also of general type and the induced action on $\Lambda_{S'}(G/E_S(G))$ is faithful.

We will use the following lemma. The proof appears in [BFGS], though the statement is different.

Lemma 2.27. [BFGS, Lemma 4.10] *Suppose that N is a normal subgroup of G , and G acts on a hyperbolic space S such that the action of N is elliptic. Then there is a G -invariant $Y \subset S$, a hyperbolic graph S' , a homomorphism $G/N \rightarrow \text{Isom } S'$, and a quasi-isometry $\varphi : Y \rightarrow S'$ that is G -equivariant.*

We apply Lemma 2.27 to $E_S(G)$. Since $Y \subset S$ is G -invariant, it contains orbits $\langle g \rangle y$ and $\langle f \rangle y$ of two elements $f, g \in \mathcal{L}_S(G)$ with disjoint fixed points, for some $y \in Y$. Since

$\varphi : Y \rightarrow S'$ is a quasi-isometry, the images of orbits $\varphi(\langle g \rangle y)$ and $\varphi(\langle f \rangle y)$ bi-infinitely diverge. Thus, $gE_S(G), fE_S(G) \in \mathcal{L}_{S'}(G/E_S(G))$ with disjoint fixed points because φ is G -equivariant and orbits of $E_S(G)$ are bounded. Therefore, the action of $G/E_S(G)$ on S' is of general type. The preimage of $E_{S'}(G/E_S(G))$ under the homomorphism $G \rightarrow G/E_S(G)$ is a normal subgroup of G and, since $\varphi : Y \rightarrow S'$ is a G -equivariant quasi-isometry, it is also elliptic. By the maximality of $E_S(G)$ (Proposition 2.24), the preimage of $E_{S'}(G/E_S(G))$ is contained in the kernel of the homomorphism. Thus, we have $E_{S'}(G/E_S(G)) = 1$ and the action of $G/E_S(G)$ on $\Lambda_{S'}(G/E_S(G))$ is faithful. $\square$

2.4 Micro-supported actions

Definition 2.28. Suppose that a group G acts on a set X . The *support of an element* $g \in G$ is the set $\text{supp}(g) = \{x \in X \mid gx \neq x\}$.

Definition 2.29. Let G be a group that acts on a set X . Given a subset $U \subseteq X$, the rigid stabilizer of U is the subgroup

$$\text{Rist}_G(U) := \{g \in G \mid gx = x \quad \forall x \in X \setminus U\}.$$

Definition 2.30. An action of a group G by homeomorphisms on a topological space X is called *micro-supported* if, for every nonempty open subset $U \subseteq X$, there is an element of G acting non-trivially on U and trivially on the complement $X \setminus U$. If the action of G on X is faithful, this is the same as having $\text{Rist}_G(U) \neq 1$ for every nonempty open subset $U \subseteq X$.

The following fact is standard and known to experts (for example, see the proof of [CRW, Proposition H]).

Lemma 2.31. *Let G be a subgroup of the homeomorphism group of a Hausdorff topological space X . If the G -action is micro-supported, then $\text{Rist}_G(U)$ is non-abelian for every nonempty open subset $U \subseteq X$.*

Lemma 2.32. *Suppose that the action of a group G on a topological space X by homeomorphisms is faithful and micro-supported. Then it is either topologically k -transitive for all $k \in \mathbb{N}$ or G satisfies a non-trivial mixed identity.*

Proof. Suppose that the action of G on X is not topologically k -transitive for some $k \in \mathbb{N}$. Then there exist nonempty open subsets $A_1, \dots, A_k$ and $B_1, \dots, B_k$ of X such that for every $g \in G$ there exists $i \in \{1, \dots, k\}$ with $A_i \cap gB_i = \emptyset$. Let us fix an arbitrary $\hat{g} \in G$ and, without loss of generality, assume that $A_1 \cap \hat{g}B_1 = \emptyset$.

Since the action of G on X is micro-supported, we can find non-trivial elements $a_1, \dots, a_k$ and $b_1, \dots, b_k \in G$, such that $\text{supp}(a_i) \subseteq A_i$ and $\text{supp}(b_i) \subseteq B_i$ for each $i \in \{1, \dots, k\}$. Then for any point $x \notin \hat{g}^{-1}A_1$ we have $\hat{g}^{-1}a_1\hat{g}x = \hat{g}^{-1}\hat{g}x = x$. Therefore, $\text{supp}(a_1^{\hat{g}}) \subseteq \hat{g}^{-1}A_1$ and we obtain $\text{supp}(a_1^{\hat{g}}) \cap \text{supp}(b_1) = \emptyset$. We conclude that $a_1^{\hat{g}}$ and b_1 commute, since they have disjoint support.

This argument shows that for an arbitrary $g \in G$ we have

$$[a_1^g, b_1] = 1 \vee [a_2^g, b_2] = 1 \vee \dots \vee [a_k^g, b_k] = 1. \quad (6)$$

Denote by $c_i := [a_i^t, b_i] \in G * \langle t \rangle$. Since $a_1, \dots, a_k$ and $b_1, \dots, b_k$ are non-trivial, the Normal Form Theorem (see, for example, [LS, Theorem 1.2, Chapter 4]) implies that each c_i is a non-trivial element of $G * \langle t \rangle$. Let $x_1, \dots, x_{k-1}$ be elements of a free group $F(x_1, \dots, x_{k-1})$, and let w_k be the iterated commutator of the elements $c_1, c_2^{x_1}, \dots, c_k^{x_{k-1}}$:

$$w_k = [c_k^{x_{k-1}}, [c_{k-2}^{x_{k-2}}, [\dots [c_1, c_2^{x_1}] \dots]].$$

Clearly, w_k is a non-trivial element of $G * F(t, x_1, x_2, \dots, x_{k-1})$. By (6), G satisfies the mixed identity $w_k = 1$. $\square$

Definition 2.33. Suppose that a group G acts on a topological space X by homeomorphisms. Let $H \leq G$. A subset V of X is called *H-compressible* if it is nonempty and if for any nonempty open subset U , there exists $h \in H$ such that $hV \subseteq U$. We say that the action of G is *compressible* if a nonempty G -compressible open subset exists and *fully compressible* if every non-dense open set is G -compressible.

Lemma 2.34. *Suppose that the action of a group G on a hyperbolic space S is quasi-parabolic, and let $\text{Fix}_{\partial S}(G) = \{q\}$. Then every nonempty open subset $V \subset \Lambda_S(G)$ with $q \notin \bar{V}$ is G -compressible. More precisely, given any nonempty open subset $V \subset \Lambda_S(G)$ with $q \notin \bar{V}$ and any nonempty open $U \subseteq \Lambda_S(G)$, there exists $l \in \mathcal{L}_S(G)$ such that $lV \subset U$. If the action of G on S is of general type, then the action of G on $\Lambda_S(G)$ is fully compressible. More precisely, given any nonempty non-dense open subset $V \subset \Lambda_S(G)$ and any nonempty open $U \subseteq \Lambda_S(G)$, there exists $l \in \mathcal{L}_S(G)$ such that $lV \subset U$.*

Proof. Let $V \subset \Lambda_S(G)$ be a nonempty open subset with $q \notin \bar{V}$. Take any nonempty open subset $U \subseteq \Lambda_S(G)$. Since $q \notin \bar{V}$, there exist two disjoint open subsets $F^+ \subset U$ and $F^- \subset \Lambda_S(G) \setminus \bar{V}$ such that $q \in F^-$. By Lemma 2.9, there exists $f \in \mathcal{L}_S(G)$ with $f^+ \in F^+$ and $f^- = q$. By Lemma 2.6 there exists $k \in \mathbb{N}$ such that $f^k(\Lambda_S(G) \setminus F^-) \subseteq F^+$. Taking $l = f^k$, we obtain

$$lV = f^k V \subset f^k(\Lambda_S(G) \setminus F^-) \subset F^+ \subset U.$$

The proof of the general type case is similar. $\square$

Lemma 2.35. *Suppose that the action of a group G on a hyperbolic space S is of general type. If the induced action on $\Lambda_S(G)$ is faithful and not topologically free, it is micro-supported.*

Proof. Let U be an arbitrary nonempty open subset of $\Lambda_S(G)$. Since the action of G on $\Lambda_S(G)$ is not topologically free, there exists $a \in G \setminus \{1\}$ such that $\text{Int}(\text{Fix}_{\Lambda_S(G)}(\langle a \rangle)) \neq \emptyset$. Therefore, $V = \Lambda_S(G) \setminus \text{Fix}_{\Lambda_S(G)}(\langle a \rangle)$ is a non-dense open subset of $\Lambda_S(G)$. Since the action of G on $\Lambda_S(G)$ is faithful, V is nonempty and $a \in \text{Rist}_G(V)$ acts nontrivially on V . Then, by Lemma 2.34, there exists $l \in \mathcal{L}_S(G)$ such that $lV \subset U$. Therefore, $l \text{Rist}_G(V) l^{-1} \leq \text{Rist}_G(U)$. $\square$

Corollary 2.36. *Let G be a faithful weakly hyperbolic group that is not lim-free. Then G contains infinite direct sums of infinite non-abelian subgroups.*

Proof. Let S be a hyperbolic space such that the action of G on S is of general type, and the induced action on $\Lambda_S(G)$ is faithful and not topologically free. Take an infinite set of pairwise disjoint non-empty open subsets $V_1, \dots, V_n, \dots \subset \Lambda_S(G)$. By Lemma 2.35, rigid stabilizers $\text{Rist}_G(V_i)$ are nonempty, and, by Lemma 2.31, they are non-abelian for all $i \in \mathbb{N}$. Elements in $\text{Rist}_G(V_i)$ and $\text{Rist}_G(V_j)$ have disjoint support for $i \neq j$, therefore they commute. Thus, $\text{Rist}_G(V_i), i \in \mathbb{N}$, form a direct sum. Every nonempty open subset of $\Lambda_S(G)$ contains infinitely many disjoint nonempty open subsets. Therefore, rigid stabilizers $\text{Rist}_G(V_i)$ are infinite. $\square$

We already know that if the G -action on a Hausdorff topological space X is micro-supported, then the rigid stabilizer of a nonempty open subset $U \subset X$ is non-abelian. If, in addition, U is G -compressible, we can say more.

Lemma 2.37. *Let G be a subgroup of the homeomorphism group of a Hausdorff topological space X , and $U \subset X$ be a nonempty open G -compressible subset. Either $\text{Rist}_G(U)$ is torsion-free, or there exists a finite $k \geq 2$ such that $\text{Rist}_G(U) \wr \mathbb{Z}_k < \text{Rist}_G(U)$.*

Proof. Suppose that a non-identity element $g \in \text{Rist}_G(U)$ has finite order. Take $x \in U$ such that the orbit of x under the action of $\langle g \rangle$ has the maximum size k . Since the action of G on X is faithful, $k \geq 2$ and divides the order of g . Consider the distinct points $gx, \dots, g^{k-1}x$. Since X is Hausdorff, there exist disjoint open subsets $V_0, \dots, V_{k-1}$, such that $g^i x \in V_i$ for $i \in \{0, \dots, k-1\}$. Consider the subset

$$V = \bigcap_{i=0}^{k-1} g^{-i} V_i.$$

It is open and nonempty since $x \in V$. By the maximality of the orbit size, a quotient of $\langle g \rangle$ isomorphic to $\mathbb{Z}^k$ acts on V by permutations, and $V, gV, \dots, g^{k-1}V$ are pairwise disjoint.

Since U is G -compressible, there exists $f \in G$ such that $fU \subset V$. Therefore,

$$g^i f \text{Rist}_G(U) (g^i f)^{-1} \leq \text{Rist}_G(g^i V) \leq \text{Rist}_G(U), \quad i \in \{0, \dots, k-1\}.$$

The subgroups $g^i f \text{Rist}_G(U) (g^i f)^{-1}$ commute, since their elements have disjoint support, and the subgroup $\langle g \rangle < \text{Rist}_G(U)$ acts on the direct product

$$\prod_{i=0}^{k-1} g^i f \text{Rist}_G(U) (g^i f)^{-1}$$

permuting the terms. Thus, $\text{Rist}_G(U) \wr \mathbb{Z}_k < \text{Rist}_G(U)$. $\square$

3 Boundary dynamics and algebraic properties

3.1 The cross-ratio and topological transitivity of actions on the limit set

In this subsection, we develop a technical tool that is later used in the proof of Proposition 3.16. We also use it to prove that the boundary action arising from a general-type action is never topologically 4-transitive (see Proposition 3.7). A similar notion in the case of proper hyperbolic spaces was considered in [Pau96].

Definition 3.1. Let a, b, c, d be four distinct points of ∂S and let $o \in S$. Then the *cross-ratio* $[a, b, c, d]_o$ is the real number defined by

$$[a, b, c, d]_o = \langle a|b \rangle_o + \langle c|d \rangle_o - \langle a|d \rangle_o - \langle b|c \rangle_o. \quad (7)$$

Since the Gromov product of distinct points of the boundary is finite and is preserved by isometries of S , we can see that the cross-ratio of distinct points of ∂S is also finite, and is also preserved by isometries of S , i.e., if g is an isometry of S , then

$$[ga, gb, gc, gd]_{go} = [a, b, c, d]_o.$$

For $x, y \in \mathbb{R}$, $\varepsilon \geq 0$, we say that $x \approx_\varepsilon y$ if $|x - y| \leq \varepsilon$.

Remark 3.2. Note that for arbitrary bounded sequences $(p_i), (q_i) \in \mathbb{R}$, if

$$\begin{aligned} \liminf_{i \rightarrow \infty} p_i \approx_\varepsilon \limsup_{i \rightarrow \infty} p_i \text{ and } \liminf_{i \rightarrow \infty} q_i \approx_\varepsilon \limsup_{i \rightarrow \infty} q_i, \text{ then} \\ \liminf_{i \rightarrow \infty} p_i + \liminf_{i \rightarrow \infty} q_i \approx_{2\varepsilon} \liminf_{i \rightarrow \infty} (p_i + q_i). \end{aligned}$$

The next lemma shows that the cross-ratio does not depend much on the choice of the base point $o \in S$.

Lemma 3.3. Let a, b, c, d be distinct points of ∂S and let $o, o' \in S$. Then

$$[a, b, c, d]_o \approx_{48\delta} [a, b, c, d]_{o'}.$$

Proof. Let $(x_i), (y_i), (z_i)$, and (t_i) be arbitrary sequences converging to a, b, c , and d , respectively. Applying Lemma 2.5(1), we get

$$[a, b, c, d]_o \approx_{8\delta} \liminf_{i \rightarrow \infty} \langle x_i|y_i \rangle_o + \liminf_{i \rightarrow \infty} \langle z_i|t_i \rangle_o - \liminf_{i \rightarrow \infty} \langle x_i|t_i \rangle_o - \liminf_{i \rightarrow \infty} \langle y_i|z_i \rangle_o. \quad (8)$$

Also, since points a, b, c, d are distinct, $\langle a|b \rangle_o, \langle c|d \rangle_o, \langle a|d \rangle_o$, and $\langle b|c \rangle_o$ are finite; therefore, sequences $(\langle x_i|y_i \rangle_o), (\langle z_i|t_i \rangle_o), (\langle x_i|t_i \rangle_o)$ and $(\langle y_i|z_i \rangle_o)$ are bounded. Thus, by (8), Lemma 2.5, Remark 3.2, and the definition of the Gromov product, we obtain

$$\begin{aligned} [a, b, c, d]_o &\approx_{24\delta} \liminf_{i \rightarrow \infty} (\langle x_i|y_i \rangle_o + \langle z_i|t_i \rangle_o - \langle x_i|t_i \rangle_o - \langle y_i|z_i \rangle_o) = \\ &= 1/2 \liminf_{i \rightarrow \infty} (d_S(x_i, t_i) + d_S(y_i, z_i) - d_S(x_i, y_i) - d_S(z_i, t_i)). \end{aligned}$$

Therefore, the cross-ratio $[a, b, c, d]_o$ is independent of the choice of $o \in S$ up to 48δ . $\square$

The next lemma is technical and will be used mainly in the proof of stability of the cross-ratio.

Lemma 3.4. *Let $o \in S$, $a, b \in \partial S$ and $\langle a|b \rangle_o = R < \infty$. Then for arbitrary $a' \in \mathcal{U}_o(a, R + 2\delta)$ and $b' \in \mathcal{U}_o(b, R + 2\delta)$, we have*

$$\max\{\langle a'|b \rangle_o, \langle a|b' \rangle_o, \langle a'|b' \rangle_o\} \leq \min\{\langle a'|a \rangle_o, \langle b'|b \rangle_o\} \quad (9)$$

Proof. Assume that $\langle a'|b \rangle_o > \langle a'|a \rangle_o$. Then by Lemma 2.5(2), we have

$$R = \langle a|b \rangle_o \geq \min\{\langle a|a' \rangle_o, \langle a'|b \rangle_o\} - \delta = \langle a|a' \rangle_o - \delta > R + \delta,$$

which gives a contradiction. The inequality $\langle a'|b \rangle_o \leq \langle b'|b \rangle_o$ and inequalities for $\langle a|b' \rangle_o$ follow by the same reasoning.

Now assume that $\langle a'|b' \rangle_o > \langle b'|b \rangle_o$. Then by Lemma 2.5(2), we have

$$\begin{aligned} R = \langle a|b \rangle_o &\geq \min\{\langle a|a' \rangle_o, \langle a'|b \rangle_o\} - \delta \geq \min\{\langle a|a' \rangle_o, \min\{\langle a'|b' \rangle_o, \langle b'|b \rangle_o\} - \delta\} - \delta = \\ &= \min\{\langle a|a' \rangle_o, \langle b'|b \rangle_o - \delta\} - \delta > R, \end{aligned}$$

which gives a contradiction. Similarly, $\langle a'|b' \rangle_o \leq \langle a'|a \rangle_o$. $\square$

Lemma 3.5. *Let $o \in S$ and $|\Lambda_S(G)| = \infty$. If $\mathcal{O}$ is a dense subset of $\Lambda_S(G)$, then there exist distinct $a, b, c, d \in \mathcal{O}$ such that $[a, b, c, d]_o$ is arbitrarily large.*

Proof. Let $M > 0$, $o \in S$, and $a, c \in \mathcal{O}$ be given. Let $\langle a|c \rangle_o = K < \infty$. Take $N = M/2 + 2(K + \delta)$. If $|\Lambda_S(G)| = \infty$, then, by Lemma 2.9, $\mathcal{H}_S(G)$ is dense in $\Lambda_S(G)$; in particular, $\Lambda_S(G)$ is Hausdorff and perfect. Since $\mathcal{O}$ is dense in $\Lambda_S(G)$, it is not discrete and has no isolated points. Thus, there exist $b \in (\mathcal{U}_o(a, N) \cap \mathcal{O}) \setminus \{a\}$ and $d \in (\mathcal{U}_o(c, N) \cap \mathcal{O}) \setminus \{c\}$. Then, by Lemma 3.4 and Lemma 2.5(2), we have

$$K = \langle a|c \rangle_o \geq \min\{\langle a|b \rangle_o, \langle b|c \rangle_o\} - \delta = \langle b|c \rangle_o - \delta.$$

Therefore, $\langle b|c \rangle_o \leq K + \delta$. Similarly, $\langle a|d \rangle_o \leq K + \delta$. Thus, we obtain

$$\langle a|b \rangle_o + \langle c|d \rangle_o - \langle a|d \rangle_o - \langle b|c \rangle_o > 2N - 2(K + \delta) = M. \quad \square$$

The next lemma shows that the cross-ratio is stable, i.e., it does not change much if we slightly move the points of the boundary.

Lemma 3.6. *Let a, b, c , and d be distinct points of ∂S , and let $o \in S$. Define $R = \max\{\langle a|b \rangle_o, \langle c|d \rangle_o, \langle a|d \rangle_o, \langle b|c \rangle_o\} + 2\delta$. If $a' \in \mathcal{U}_o(a, R)$, $b' \in \mathcal{U}_o(b, R)$, $c' \in \mathcal{U}_o(c, R)$, and $d' \in \mathcal{U}_o(d, R)$, then*

$$[a, b, c, d]_o \approx_{8\delta} [a', b', c', d']_o.$$

Proof. By Lemma 2.5(2), we have

$$\langle a'|b' \rangle_o \geq \min \{ \langle a'|a \rangle_o, \min \{ \langle a|b \rangle_o, \langle b|b' \rangle_o \} - \delta \} - \delta = \langle a|b \rangle_o - 2\delta.$$

Similarly, by Lemma 2.5(2) and Lemma 3.4, we obtain $\langle a|b \rangle_o \geq \langle a'|b' \rangle_o - 2\delta$, and thus, $\langle a|b \rangle_o \approx_{2\delta} \langle a'|b' \rangle_o$. Using the same argument, we get

$$\langle c|d \rangle_o \approx_{2\delta} \langle c'|d' \rangle_o, \langle a|d \rangle_o \approx_{2\delta} \langle a'|d' \rangle_o, \text{ and } \langle b|c \rangle_o \approx_{2\delta} \langle b'|c' \rangle_o.$$

This gives us the desired inequality $|[a, b, c, d]_o - [a', b', c', d']_o| \leq 8\delta$. $\square$

Proposition 3.7. *Let G be a group admitting a non-elementary action on a hyperbolic space S , and let $\mathcal{O}$ be a G -invariant subset of $\Lambda_S(G)$.*

- (a) *If the action of G on S is of general type, then the induced action of G on $\mathcal{O}$ is not topologically 4-transitive.*
- (b) *If the action of G on S is quasi-parabolic, then the induced action of G on $\mathcal{O}$ is not topologically 3-transitive.*

Proof. By Lemma 2.9, the set of fixed points of elements of $\mathcal{L}_S(G)$ is dense in $\Lambda_S(G)$. Thus, since $\mathcal{O}$ is G -invariant, by Lemma 2.6, it is also dense in $\Lambda_S(G)$.

- (a) Let $o \in S$. By Lemma 3.5, the cross-ratio of distinct points of $\mathcal{O}$ can be arbitrarily large. Therefore, there exist eight distinct points $a, b, c, d, a', b', c', d' \in \mathcal{O}$ such that

$$|[a, b, c, d]_o - [a', b', c', d']_o| \geq 65\delta.$$

Define

$$R_1 = \max \{ \langle a|b \rangle_o, \langle c|d \rangle_o, \langle a|d \rangle_o, \langle b|c \rangle_o \} + 2\delta \text{ and}$$

$$R_2 = \max \{ \langle a'|b' \rangle_o, \langle c'|d' \rangle_o, \langle a'|d' \rangle_o, \langle b'|c' \rangle_o \} + 2\delta.$$

By way of contradiction, assume that the action of G on $\mathcal{O}$ is topologically 4-transitive. Then there exists $g \in G$ and points

$$\widehat{a} \in \mathcal{U}_o(a', R_2) \cap \mathcal{O}, \widehat{b} \in \mathcal{U}_o(b', R_2) \cap \mathcal{O}, \widehat{c} \in \mathcal{U}_o(c', R_2) \cap \mathcal{O}, \widehat{d} \in \mathcal{U}_o(d', R_2) \cap \mathcal{O}$$

such that

$$g\widehat{a} \in \mathcal{U}_o(a, R_1) \cap \mathcal{O}, g\widehat{b} \in \mathcal{U}_o(b, R_1) \cap \mathcal{O}, g\widehat{c} \in \mathcal{U}_o(c, R_1) \cap \mathcal{O}, \text{ and } g\widehat{d} \in \mathcal{U}_o(d, R_1) \cap \mathcal{O}.$$

Using Lemma 3.6 and Lemma 3.3, we obtain

$$\begin{aligned} 65\delta &\leq |[a, b, c, d]_o - [a', b', c', d']_o| \leq |[a, b, c, d]_o - [\widehat{a}, \widehat{b}, \widehat{c}, \widehat{d}]_o| + |[\widehat{a}, \widehat{b}, \widehat{c}, \widehat{d}]_o - [a', b', c', d']_o| \leq \\ &\leq |[a, b, c, d]_o - [g\widehat{a}, g\widehat{b}, g\widehat{c}, g\widehat{d}]_o| + |[g\widehat{a}, g\widehat{b}, g\widehat{c}, g\widehat{d}]_o - [g\widehat{a}, g\widehat{b}, g\widehat{c}, g\widehat{d}]_{go}| + \\ &+ |[\widehat{a}, \widehat{b}, \widehat{c}, \widehat{d}]_o - [a', b', c', d']_o| \leq 64\delta, \end{aligned}$$

which is a contradiction.

- (b) Let q be a global fixed point in $\Lambda_S(G)$. Let $o \in S$. Again, using Lemma 3.5, we find distinct points $a, b, c, a', b', c' \in \mathcal{O}$ such that

$$|[a, b, c, q]_o - [a', b', c', q]_o| \geq 65\delta.$$

The rest of the proof is similar to part (a), except that we use a fixed point q instead of d and d' .

□

Remark 3.8. M. Gromov [Gro87, 8.2.H] observed that if a group G admits a non-elementary action on a hyperbolic space S , then the induced action of G on $\Lambda_S(G)$ is always topologically transitive. If, in addition, the action of G on S is of general type, then the induced action on $\Lambda_S(G)$ is always topologically 2-transitive.

We will later use the following easy observation.

Lemma 3.9. *If an action of a group G on a dense subset $\mathcal{O}$ of a Hausdorff topological space X without isolated points is k -transitive, then it is topologically k -transitive.*

Proof. Let $U_1, \dots, U_k$ and $V_1, \dots, V_k$ be nonempty open subsets of $\mathcal{O}$. Being a dense subset of a Hausdorff topological space without isolated points, $\mathcal{O}$ is itself Hausdorff without isolated points. Therefore, we can find pairwise distinct $x_1, \dots, x_k$ and pairwise distinct $y_1, \dots, y_k$ such that $x_i \in U_i$ and $y_i \in V_i$ for every $i = 1, \dots, k$. Since the action of G on $\mathcal{O}$ is k -transitive, there exists $g \in G$ such that $gy_i = x_i$ for all $i = 1, \dots, k$; in particular, $gV_i \cap U_i \neq \emptyset$ for every $i = 1, \dots, k$. □

3.2 Boundary dynamics and mixed identities

Theorem 3.10. *Suppose that an action of a group G on a hyperbolic space S is of general type and the induced action on $\Lambda_S(G)$ is faithful. Then the following are equivalent:*

- (1) G is MIF.
- (2) The induced action of G on $\Lambda_S(G)$ is topologically free.

Proof. (1) $\Rightarrow$ (2). By contrapositive, suppose that the action of G on $\Lambda_S(G)$ is not topologically free. Then by Lemma 2.35, it is micro-supported. The action of G on $\Lambda_S(G)$ is not topologically 4-transitive by Proposition 3.7. Therefore, by Lemma 2.32, G satisfies a nontrivial mixed identity.

(2) $\Rightarrow$ (1). Assume that the action of G on $\Lambda_S(G)$ is topologically free. Since the action of G on S is of general type, G contains a free subgroup and does not satisfy any nontrivial identity. By [HO16, Rem. 5.1], G satisfies some non-trivial mixed identity $w \in G * F_n$ if and only if it satisfies a non-trivial mixed identity $w' \in G * \langle t \rangle$. Thus, it is sufficient to prove that G does not satisfy any mixed identity $w \in (G * \langle t \rangle) \setminus \langle t \rangle$.

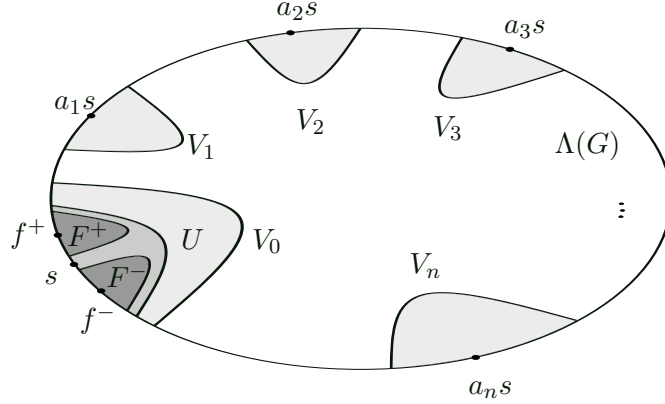


Figure 5: For Theorem 3.10

Take any reduced nonempty word $w \in (G * \langle t \rangle) \setminus \langle t \rangle$. Up to conjugation $w = t^{n_1} a_1 t^{n_2} a_2 \dots a_{m-1} t^{n_m} a_m$, where $a_i \in G \setminus \{1\}$, $n_i \in \mathbb{Z} \setminus \{0\}$, $i \in \{1, \dots, m\}$.

Since G acts on $\Lambda_S(G)$ by homeomorphisms, and the action is topologically free, the sets $\text{Fix}_{\Lambda_S(G)}(\langle a_i \rangle)$ are closed in $\Lambda_S(G)$ and $\text{Int}(\text{Fix}_{\Lambda_S(G)}(\langle a_i \rangle)) = \emptyset$ for every $i \in \{1, \dots, m\}$. Let

$$M = \bigcap_{i \in \{1 \dots m\}} \Lambda_S(G) \setminus \text{Fix}_{\Lambda_S(G)}(\langle a_i \rangle).$$

The set M is a finite intersection of open dense sets, so it is open and dense; in particular, it is nonempty, and there exists $s \in \Lambda_S(G)$ such that $a_i s \neq s$ for every $i \in \{1, \dots, m\}$.

Since $\Lambda_S(G)$ is Hausdorff, there exist open subsets V_0 and $V_1, \dots, V_m$ of $\Lambda_S(G)$ such that $s \in V_0$, $a_i s \in V_i$ and $V_0 \cap V_i = \emptyset$ for every $i \in \{1, \dots, m\}$. Take $U = V_0 \cap a_1^{-1}(V_1) \cap \dots \cap a_m^{-1}(V_m)$ to ensure that $U \cap a_i(U) = \emptyset$ for all $i \in \{1, \dots, m\}$.

The action of G on S is of general type and $U \setminus \{s\}$ is open; therefore, by Lemma 2.9, there exists $f \in \mathcal{L}_S(G)$ such that $\{f^+, f^-\} \subset U \setminus \{s\}$. Let F^+ and F^- be two disjoint open subsets of $U \setminus \{s\}$ such that $f^+ \in F^+$ and $f^- \in F^-$. By Lemma 2.6 there exists $n \in \mathbb{N}$ such that

$$f^n(\Lambda_S(G) \setminus F^-) \subseteq F^+ \text{ and } f^{-n}(\Lambda_S(G) \setminus F^+) \subseteq F^-.$$

Take $g = f^n$. We have

$$\begin{aligned} w(g)s &= g^{n_1} a_1 \dots a_{m-1} g^{n_m} a_m s \in g^{n_1} a_1 \dots a_{m-1} g^{n_m} a_m(U) \subset g^{n_1} a_1 \dots a_{m-1} g^{n_m}(\partial S \setminus U) \subset \\ &\subset g^{n_1} a_1 \dots a_{m-1}(F^+ \cup F^-) \subset g^{n_1} a_1 g^{n_2} a_2 \dots a_{m-1}(U) \subset \dots \subset g^{n_1} a_1(U) \subset F^+ \cup F^-. \end{aligned}$$

Since $s \notin F^+ \cup F^-$, we obtain that $w(g)s \neq s$; in particular, $w(g) \neq 1$ in G . $\square$

Proof of Theorem B. Condition (b) obviously implies (a). The equivalence of conditions (a) and (c) is the statement of Theorem 3.10. Being MIF is a purely algebraic property of a group. So, by Theorem 3.10, if G is MIF, then any general type action on any hyperbolic

space S should induce a topologically free action on $\Lambda_S(G)$ whenever the action on $\Lambda_S(G)$ is faithful. By the definition of an elliptic radical, the induced action of G on $\Lambda_S(G)$ is faithful if and only if $E_S(G)$ is trivial. Therefore, (c) implies (b). $\square$

Proof of Corollary C. Let H be a proper normal subgroup of G . Then, by Lemma 2.26, H is itself lim-free, and $\Lambda_S(H) = \Lambda_S(G)$. Suppose that H is algebraic. Then there exists a finite collection of mixed identities $\{w_1(t), \dots, w_n(t)\}$ in $G*\langle t \rangle$ such that for any $h \in H$ there exists $i \in \{1, \dots, n\}$ such that $w_i(h) = 1$ in G . Up to conjugation each $w_i(t)$ is either simply t^m for some $m \in \mathbb{N}$ or of the form $w_i(t) = t^{n_1}a_1t^{n_2}a_2 \dots a_{m-1}t^{n_m}a_m$, where $a_j \in G \setminus \{1\}$, $n_j \in \mathbb{Z} \setminus \{0\}$, $j \in \{1, \dots, m\}$. Let $\{a_1, \dots, a_k\}$ be the finite collection of all coefficients from G appearing in the mixed identities $\{w_1(t), \dots, w_n(t)\}$. Repeating the argument of Theorem 3.10, we can find $s \in \Lambda_S(H)$, such that $a_i s \neq s$ for every $i \in \{1, \dots, k\}$, and $h \in \mathcal{L}_S(H)$ such that $w_i(h)s \neq s$ simultaneously for all $w_i(t)$ containing nontrivial coefficients from G . Note that $h^m \neq 1$ in G for every $m \in \mathbb{N}$, since h is loxodromic. In particular, $w_i(h) \neq 1$ for all $i = 1, \dots, n$ in G , which contradicts that H is algebraic. $\square$

Now, we show that faithful weakly hyperbolic groups that are not lim-free also have an interesting algebraic property.

Proof of Proposition D. Let S be a hyperbolic space such that the action of G on S is of general type, and the induced action of G on $\Lambda_S(G)$ is faithful and not topologically free. Let $\mathcal{C}$ be the collection of all nonempty non-dense open subsets of $\Lambda_S(G)$. Let $H = \langle \text{Rist}_G(Y) \mid Y \in \mathcal{C} \rangle$. By Lemma 2.35, the action of G on $\Lambda_S(G)$ is micro-supported; in particular, H is a nontrivial normal subgroup of G . Since the action of G on $\Lambda_S(G)$ is faithful, H acts on $\Lambda_S(G)$ nontrivially. Therefore, by Lemma 2.10, the action of H on S is also of general type and $\Lambda_S(H) = \Lambda_S(G)$. Thus, by Lemma 2.34, the action of H on $\Lambda_S(G)$ is fully compressible. In particular, $\mathcal{C}$ satisfies the conditions of [GR, Corollary 1.7], and $\text{Mon}(G) = [H, H]$ is non-abelian and simple. Assume $\text{Mon}(G)$ is finite. Then $\text{Mon}(G)$ is a normal elliptic subgroup of G , and, by Proposition 2.24, $\text{Mon}(G) \leq E_S(G)$. In particular, $E_S(G) \neq 1$, which contradicts that the action of G on $\Lambda_S(G)$ is faithful. $\square$

Proposition D does not apply to any acylindrically hyperbolic group. In fact, every acylindrically hyperbolic group has a trivial monolith.

Proposition 3.11. *Let G be an acylindrically hyperbolic group. Then $\text{Mon}(G) = 1$.*

Proof. By [DGO, Thm 2.24 (c)], there exists a hyperbolically embedded subgroup $H \leq G$ such that $H \cong \mathbb{Z} \times K(G)$, where $K(G)$ is the unique maximal finite normal subgroup of G . Let g be any nontrivial element of G . Then, by [DGO, Thm 7.19], for the finite subset $Z = \{g, 1\}$ there exists a finite subset $F \subseteq H \setminus \{1\}$ such that for every normal subgroup N of H , satisfying $N \cap F = \emptyset$, the natural epimorphism $\varepsilon : G \rightarrow \overline{G}$ is injective on Z , where

$$\overline{G} = G / \langle\langle N \rangle\rangle^G.$$

Since F is finite and $H \cong \mathbb{Z} \times K(G)$, a nontrivial normal subgroup N with the property $N \cap F = \emptyset$ always exists. Thus, $\overline{G}$ is a proper quotient, in which the image of g is nontrivial; therefore, the kernel of the natural homomorphism avoids g . Since g was an arbitrary nontrivial element of G , $\text{Mon}(G) = 1$. $\square$

This allows us to obtain the result of D. Osin and M. Hull [HO16, Cor. 5.10] independently, as a corollary of Proposition 3.11, Theorem B, and Proposition D.

Corollary 3.12. *If the action of a group G on a hyperbolic space S is of general type and acylindrical, then the following conditions are equivalent*

1. *The finite radical $K(G)$ is trivial.*
2. *The action of G on $\Lambda_S(G)$ is faithful.*
3. *The action of G on $\Lambda_S(G)$ is topologically free.*
4. *G is MIF.*

Proof. The equivalence of (1) and (2) follows from $K(G) = E_S(G)$, by Lemma 2.25. To obtain (2) $\implies$ (3), note that by Proposition 3.11, $\text{Mon}(G) = 1$, and by Proposition D, G is lim-free. (3) $\iff$ (4) is Theorem B, and (4) $\implies$ (1) is obvious. $\square$

Proposition D also gives us another large class of lim-free weakly hyperbolic groups.

Proof of Corollary E. If a faithful weakly hyperbolic residually finite group G is not lim-free, then, by Proposition D, $\text{Mon}(G)$ is infinite and simple. This contradicts the fact that every subgroup of a residually finite group is residually finite. By a well-known result of Malcev [Mal], every finitely generated linear group is residually finite. Thus, every finitely generated faithful weakly hyperbolic linear group is lim-free. By Theorem B, every faithful weakly hyperbolic residually finite group is MIF. $\square$

In the special case of subgroups of $\text{PSL}_2(\mathbb{C})$, we do not need finite generation.

Proof of Corollary F. Let $\mathbb{H}^3 = \{z + tj : z \in \mathbb{C}, t > 0\}$ be the upper half-space model of hyperbolic three-space.

$$\text{PSL}_2(\mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| a, b, c, d \in \mathbb{C}, ad - bc = 1 \right\} / \{\pm I\}.$$

Every subgroup H of $\text{PSL}_2(\mathbb{C})$ acts on $\mathbb{H}^3$ by Möbius transformations:

$$g(z) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} z = \frac{az + b}{cz + d}, \quad z \in \mathbb{C},$$

$$g(\infty) = \infty \text{ if } c = 0 \text{ and } g\left(\frac{-d}{c}\right) = \infty, \quad g(\infty) = \frac{a}{c} \text{ if } c \neq 0.$$

See [Bea, Chapter 4] for more details. The boundary of $\mathbb{H}^3$ is $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. Analyzing solutions of the equation $cz^2 + (d - a)z - b = 0$, where $z \in \Lambda_{\mathbb{H}^3}(H)$, we obtain that an element of H has more than two fixed points in $\Lambda_{\mathbb{H}^3}(H)$ if and only if $c = b = 0$ and $d = a$, which happens only when it is trivial. Thus, the action of H on $\Lambda_{\mathbb{H}^3}(H)$ is topologically free. Therefore, by Theorem B, if the action of H on $\mathbb{H}^3$ is of general type, then H is MIF.

Assume that the action of H on $\mathbb{H}^3$ is not of general type. Then either H fixes a point on $\partial\mathbb{H}^3$ or the action of H is elliptic or lineal, exchanging the two points in the limit set.

If the action of H is elliptic, then H fixes a point in $\mathbb{H}^3$ [Bea, Thm 4.3.7]. Therefore, H is conjugate to a subgroup of $\mathrm{SO}(3)$. Every proper closed (in the Euclidean topology) subgroup of $\mathrm{SO}(3)$ is virtually abelian. Therefore, if H is not virtually solvable, then H is dense in $\mathrm{SO}(3)$. Therefore, H is Zariski dense, and, by the result of Tomanov [Tom], is MIF.

If the action of H is lineal, exchanging the two points in the limit set, then H contains a subgroup N of index 2, fixing these points in $\partial\mathbb{H}^3$.

Now let us consider the case in which H fixes a point in $\partial\mathbb{H}^3$. Since the action of $\mathrm{PSL}_2(\mathbb{C})$ is transitive on $\widehat{\mathbb{C}}$, we can always find an element $g \in \mathrm{PSL}_2(\mathbb{C})$ such that $H' = g^{-1}Hg$ fixes ∞ . Then H' consists of the upper-triangular matrices

$$\left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mid a, b, d \in \mathbb{C}, ad = 1 \right\}$$

and is solvable. Therefore, H is also solvable. □

Remark 3.13. Unfortunately, Corollary E does not generalize to arbitrary finitely generated weakly hyperbolic linear groups; the faithfulness condition is essential here. Neither does Corollary F generalize to subgroups of projective linear groups of higher dimension. For example, $F_2 \times \mathbb{Z} \leq \mathrm{SL}_3(\mathbb{R})$ is not virtually solvable, but also is not MIF.

The next corollary shows that results about mixed identities from [FMMS] can be obtained as a special case of Theorem 3.10.

Corollary 3.14. [FMMS, Theorem B] *Let T be a simplicial tree. Suppose that the action of G on T is faithful, minimal, and of general type. Then the following are equivalent:*

1. G is MIF;
2. The action of G on ∂T is topologically free.

Proof. To obtain the statement as a corollary of Theorem 3.10 we need to show that if the action of G on T is faithful, minimal, and of general type, then the induced action on $\Lambda_T(G)$ is also faithful. By Lemma 2.21 $\Lambda_T(G) = \partial T$. By taking the barycentric subdivision, we can assume that the action of G on T is without inversions. Therefore, by Lemma 2.22, $E_T(G)$ fixes a vertex $t \in T$. Since $E_T(G)$ is normal in G , it fixes every vertex of the orbit Gt . Thus, it stabilizes a minimal invariant subtree T_{Gt} containing this orbit. Since the action

of G on T is minimal, $T_{Gt} = T$, and, since the action of G on T is faithful, $E_T(G) = 1$. Therefore, the induced action of G on ∂T is also faithful. $\square$

We finish this section by proving Corollary [G](#), connecting our research to reduced C^* -algebras. This follows immediately from the following theorem, which is the combination of Proposition 15 and Theorem 14 from [\[Oza\]](#).

Theorem 3.15. *Let G be a group. Assume that there is a continuous action of G on a Hausdorff topological space that is minimal, fully compressible, and topologically free. Then the reduced group C^* -algebra $C_\lambda^*(G)$ is selfless.*

Proof of Corollary [G](#). Suppose a group G admits a general type action on a hyperbolic space S such that the induced action on $\Lambda_S(G)$ is topologically free. Then, by Lemma [2.34](#) and Corollary [2.8](#), the action of G on $\Lambda_S(G)$ is fully compressible and minimal. Therefore, by Theorem [3.15](#), the reduced group C^* -algebra $C_\lambda^*(G)$ is selfless. $\square$

3.3 Boundary dynamics and triple transitivity

The goal of this subsection is to prove Theorem [I](#). The proof strategy closely follows [\[BM\]](#), but the details are different. The key ingredient of the proof is the following proposition.

Proposition 3.16. *Suppose that the action of a group G on a hyperbolic space S is of general type. Let Ω be a set on which G acts faithfully and 2-transitively. Then for any $\omega \in \Omega$, the action of a subgroup G_ω on S is non-elementary, and $\Lambda_S(G_\omega) = \Lambda_S(G)$.*

We will need the following known and easily established fact (see, for example, [\[La, ex. 47\]](#)). We provide the proof to make the paper self-contained.

Lemma 3.17. *If the action of a group G on a set Ω is 2-transitive, then for any $\omega \in \Omega$, there exists an element $g \in G$ such that G admits a decomposition into double cosets $G = G_\omega \sqcup G_\omega g G_\omega$.*

Proof. Fix $\omega \in \Omega$. Take any $\omega_1, \omega_2 \in \Omega \setminus \{\omega\}$. Since the action of G on Ω is 2-transitive, the action of G_ω is transitive on $\Omega \setminus \{\omega\}$. Therefore, there exists $\gamma \in G_\omega$ such that $\gamma\omega_1 = \omega_2$. Since G acts transitively on Ω , there exists $g \notin G_\omega$ such that $g\omega = \omega_1$. Hence, $\omega_2 = \gamma g \omega$. Since ω_2 was arbitrary, we obtain $\Omega \setminus \{\omega\} = G_\omega g \omega$. Thus, if $f \in G \setminus G_\omega$, then $f\omega = \gamma_1 g \omega$ for some $\gamma_1 \in G_\omega$. So, $f = \gamma_1 g \gamma_2$ for some $\gamma_2 \in G_\omega$. Therefore, $G = G_\omega \sqcup G_\omega g G_\omega$. $\square$

Remark 3.18. Note that the subgroup $G_\omega \leq G$ in this case is maximal.

Proof of Proposition [3.16](#). Fix an arbitrary $\omega \in \Omega$. The action of G on Ω is 2-transitive; thus, by Lemma [3.17](#), there exists $g \in G$ such that $G = G_\omega \sqcup G_\omega g G_\omega$. We argue by contradiction and assume that $\Lambda_S(G_\omega) \neq \Lambda_S(G)$.

Case 1. Assume that $\Lambda_S(G_\omega) = \emptyset$, so the action of G_ω on S is elliptic. Take any point $s \in S$. Let $K = \sup\{d_S(ts, s) \mid t \in G_\omega\}$, and $M = d_S(gs, s)$. Let $f \in G$ be arbitrary. If $f \in G_\omega$, then $d_S(fs, s) \leq K$. If $f \notin G_\omega$, then, by Lemma 3.17, there exist $\gamma_1, \gamma_2 \in G_\omega$, such that $f = \gamma_1 g \gamma_2$. Thus, we have

$$d_S(fs, s) = d_S(\gamma_1 g \gamma_2 s, s) \leq d_S(g \gamma_2 s, gs) + d_S(gs, s) + d_S(s, \gamma_1^{-1} s) \leq 2K + M.$$

Therefore, G also acts on S with bounded orbits, which contradicts that the action of G on S is of general type.

Case 2. Assume that $|\Lambda_S(G_\omega)| = 1$, so the action of G_ω on S is parabolic. Let $q(\omega) = \Lambda_S(G_\omega)$. Take $l \in \mathcal{L}_S(G)$ with sufficiently large asymptotic translation length $\tau_S(l)$ and fixed points $\{l^+, l^-\}$ disjoint from $q(\omega)$. Let L_l be the standard $(2, \delta)$ -quasi-geodesic axis of l , provided by Lemma 2.15. Let T be the translation distance of l along L_l . Let N be a constant from Lemma 2.19, and let $m, n \geq N$ be such that $m \gg 4n$. Since G_ω contains no loxodromic elements, by Lemma 3.17, l^m and l^n are in the same double coset of G_ω , and there exist $\gamma_1, \gamma_2 \in G_\omega$ such that $l^m = \gamma_1 l^n \gamma_2$.

By Proposition 2.17, there exists $s \in S$ such that

$$d_S(s, \gamma_1 s) \leq 56\delta \text{ and } d_S(s, \gamma_2 s) \leq 56\delta.$$

We obtain the following:

$$\begin{aligned} d_S(l^m s, s) &= d_S(\gamma_1 l^n \gamma_2 s, s) \leq d_S(s, \gamma_1 s) + d_S(\gamma_1 s, \gamma_1 l^n \gamma_2 s) \leq \\ &\leq d_S(s, \gamma_1 s) + d_S(s, l^n s) + d_S(l^n s, l^n \gamma_2 s) \leq d_S(s, l^n s) + 112\delta. \end{aligned} \quad (10)$$

By Lemma 2.19, there exists $D(\delta) \in \mathbb{N}$ such that

$$2d_S(s, L_l) + \frac{m}{2}T - D \leq d_S(s, l^m s) \text{ and } d_S(s, l^n s) \leq 2d_S(s, L_l) + 2nT + D. \quad (11)$$

Combining (10) and (11), we get

$$mT \leq 4nT + 4(D + 56\delta),$$

which contradicts that $m \gg 4n$ and $T > 0$.

Case 3. Now, assume that $|\Lambda_S(G_\omega)| > 1$.

For an element $f \in G$ and a point $o \in S$, define

$$[\Lambda_S(G_\omega), f\Lambda_S(G_\omega)]_o := \inf\{[a, b, c, d]_o \mid a, b \in \Lambda_S(G_\omega), c, d \in f\Lambda_S(G_\omega)\},$$

if such points can be chosen pairwise distinct, so that $[a, b, c, d]_o$ is well-defined; and $[\Lambda_S(G_\omega), f\Lambda_S(G_\omega)]_o := 0$, otherwise.

Let $f \in G$ be arbitrary. By Lemma 3.17, there exists $\gamma_1, \gamma_2 \in G_\omega$ such that $f = \gamma_1 g \gamma_2$. Therefore,

$$[\Lambda_S(G_\omega), f\Lambda_S(G_\omega)]_o = [\Lambda_S(G_\omega), \gamma_1 g \gamma_2 \Lambda_S(G_\omega)]_o = [\gamma_1^{-1} \Lambda_S(G_\omega), g \gamma_2 \Lambda_S(G_\omega)]_{\gamma_1^{-1} o}.$$

Thus, since γ_1 and γ_2 preserve $\Lambda_S(G_\omega)$ setwise, using Lemma 3.3, we obtain

$$\begin{aligned} & |[\Lambda_S(G_\omega), f\Lambda_S(G_\omega)]_o - [\Lambda_S(G_\omega), g\Lambda_S(G_\omega)]_o| \\ & \leq |[\gamma_1^{-1}\Lambda_S(G_\omega), g\gamma_2\Lambda_S(G_\omega)]_o - [\Lambda_S(G_\omega), g\Lambda_S(G_\omega)]_o| + 48\delta \leq 48\delta. \end{aligned}$$

In particular, we have

$$\sup\{[\Lambda_S(G_\omega), f\Lambda_S(G_\omega)]_o \mid f \in G\} < \infty. \quad (12)$$

The subset $U = \Lambda_S(G) \setminus \Lambda_S(G_\omega)$ is nonempty and open in $\Lambda_S(G)$. Therefore, by Lemma 2.9, there exists $l \in \mathcal{L}_S(G)$ such that $\{l^+, l^-\} \subset U$. There exists $N \in \mathbb{N}$ such that $V_N = \mathcal{U}_o(l^+, N) \cap \Lambda_S(G) = \{x \in \Lambda_S(G) \mid \langle x|l^+ \rangle_o > N\} \subset U$. There exists $M \gg N$ such that $V_M = \mathcal{U}_o(l^+, M) \cap \Lambda_S(G) \subset V_N \subset U$. By Lemma 2.6 there exists $m \in \mathbb{N}$ such that $l^m\Lambda_S(G_\omega) \subset V_M$. Take $f = l^m$. Then, using Lemma 2.5, for arbitrary $c, d \in f\Lambda_S(G_\omega)$, we obtain

$$\langle c|d \rangle_o \geq \min\{\langle c|l^+ \rangle_o, \langle l^+|d \rangle_o\} - \delta \geq M - \delta. \quad (13)$$

Also, for arbitrary $a, b \in \Lambda_S(G_\omega)$, we have

$$N > \langle a|l^+ \rangle_o \geq \min\{\langle a|d \rangle_o, \langle d|l^+ \rangle_o\} - \delta, \text{ and } N > \langle b|l^+ \rangle_o \geq \min\{\langle b|c \rangle_o, \langle c|l^+ \rangle_o\} - \delta.$$

Note that $\min\{\langle a|d \rangle_o, \langle d|l^+ \rangle_o\} = \langle a|d \rangle_o$, and $\min\{\langle b|c \rangle_o, \langle c|l^+ \rangle_o\} = \langle b|c \rangle_o$ since $M \gg N$. Thus, we obtain

$$\langle a|d \rangle_o < N + \delta \text{ and } \langle b|c \rangle_o < N + \delta. \quad (14)$$

Combining (13) and (14), we get $[\Lambda_S(G_\omega), f\Lambda_S(G_\omega)]_o > M - 2N - 3\delta$, which contradicts (12), since M was arbitrarily large. Therefore, $\Lambda_S(G_\omega) = \Lambda_S(G)$. The action of G_ω on S is non-elementary, since $|\Lambda_S(G_\omega)| = |\Lambda_S(G)| = \infty$. $\square$

From now on, we fix an action of a group G on a hyperbolic space S that is of general type, and the induced action on $\Lambda_S(G)$ is faithful and not topologically free. Actions of subgroups of G on S are induced by this action. We assume that G acts faithfully and 3-transitively on a set Ω . Note that G is infinite, and thus Ω is also infinite, since the action of G on Ω is faithful.

We want to show that for any $\omega \in \Omega$, the action of G_ω on S is quasi-parabolic. Note that by the transitivity of G on Ω , all subgroups G_ω , $\omega \in \Omega$ are conjugate, and thus their actions on S are of the same type. Proposition 3.16 shows that it is either quasi-parabolic or of general type. Arguing by contradiction, we assume that the action of G_ω is of general type. The action of G_ω on $\Omega \setminus \{\omega\}$ is 2-transitive. Therefore, we can apply Proposition 3.16 to point stabilizers of points of $\Omega \setminus \{\omega\}$ in G_ω to conclude that they also induce a non-elementary action on S . This gives us two possibilities to exclude: the action of such point stabilizers on S is of general type, or it is quasi-parabolic. The former possibility will be excluded in Proposition 3.19, the latter in Proposition 3.20.

Denote by $\Omega^{(2)}$ the set of ordered pairs of elements of Ω minus the diagonal, i.e.,

$$\Omega^{(2)} = \Omega \times \Omega \setminus \{(\omega, \omega) \mid \omega \in \Omega\}.$$

Proposition 3.19. *There does not exist a set Ω such that the action of G on Ω is faithful, 3-transitive, the action of G_ω on S is of general type for every $\omega \in \Omega$, and for every $\Delta \in \Omega^{(2)}$, the action of G_Δ on S is also of general type.*

Proof. For the sake of contradiction, assume that the action of G_Δ on S is of general type for some $\Delta \in \Omega^{(2)}$. Since the action of G on $\Omega^{(2)}$ is transitive, all subgroups G_Δ are conjugate. Therefore, it is sufficient to arrive at a contradiction for a specifically chosen $\Delta \in \Omega^{(2)}$. We will explain the choice of Δ below.

By Proposition 3.7, the action of G on $\Lambda_S(G)$ is not topologically 4-transitive, therefore there exist nonempty open subsets $V_1, V_2, V_3, V_4 \subset \Lambda_S(G)$ and $W_1, W_2, W_3, W_4 \subset \Lambda_S(G)$ such that there does not exist any $f \in G_\Delta$ such that

$$W_i \cap f(V_i) \neq \emptyset \text{ for all } i = 1, 2, 3, 4.$$

Thus, setting $V_j = W_{j-4}$ for $j = 5, 6, 7, 8$, we obtain the list of nonempty open subsets $V_1, \dots, V_8$ with the property that for every $f \in G_\Delta$ there is a pair $i, j \in \{1, \dots, 8\}$ such that

$$V_i \cap fV_j = \emptyset. \quad (15)$$

By taking smaller subsets if necessary, we can assume V_i to be pairwise disjoint and not dense in $\Lambda_S(G)$ for all $i \in \{1, \dots, 8\}$.

Since the action of G on $\Lambda_S(G)$ is faithful and not topologically free, by Lemma 2.35, the action of G on $\Lambda_S(G)$ is micro-supported. Therefore, there exists a nontrivial element $g_1 \in G$ such that $\text{supp}(g_1) \subseteq V_1$. By Lemma 2.31, $\text{Rist}_G(V_1)$ cannot have all its elements of order two. Thus, we may assume that the order of g_1 is greater than two, and, since the action of G on Ω is faithful, g_1 has an orbit of size at least 3 in Ω . Let $\omega_1, \omega_2, \omega_3 \in \Omega$ be distinct points such that

$$\omega_2 = g_1\omega_1, \text{ and } \omega_3 = g_1\omega_2. \quad (16)$$

Define $\Delta = (\omega_1, \omega_2)$.

Since the action of G_Δ on S is of general type, by Lemma 2.34, we can find $u_2, u_3, u_4, u_5, u_6, u_7, u_8 \in G_\Delta$ such that $g_i = u_i g_1 u_i^{-1}$ have supports in V_i , respectively, $i = 2, \dots, 8$.

Using (16), the fact that each u_i stabilizes both ω_1 and ω_2 , and that $[g_i, g_j] = 1$ for $i, j \in \{1, \dots, 8\}$, $i \neq j$, since they have disjoint support, for all $i = 2, \dots, 8$ we obtain

$$\begin{aligned} g_i\omega_1 &= u_i g_1 u_i^{-1} \omega_1 = u_i g_1 \omega_1 = u_i \omega_2 = \omega_2, \text{ and} \\ g_i\omega_2 &= g_i g_1 \omega_1 = g_1 g_i \omega_1 = g_1 \omega_2 = \omega_3. \end{aligned} \quad (17)$$

Take any $f \in G_\Delta$. By (15), elements g_i^f and g_j have disjoint support for at least one pair $i, j \in \{1, \dots, 8\}$ and, thus, $[g_i^f, g_j] = 1$. Using this, (17), and the fact that f fixes ω_1 and ω_2 , we obtain

$$\begin{aligned} g_i^f \omega_1 &= f^{-1} g_i \omega_1 = f^{-1} \omega_2 = \omega_2 \text{ and} \\ g_i^f \omega_2 &= g_i^f g_j \omega_1 = g_j g_i^f \omega_1 = g_j \omega_2 = \omega_3. \end{aligned} \quad (18)$$

Combining (18) and $g_i^f \omega_2 = f^{-1} g_i \omega_2 = f^{-1} \omega_3$, we obtain $f \omega_3 = \omega_3$, which contradicts the assumption that the action of G on Ω is 3-transitive. $\square$

Proposition 3.20. *There does not exist a set Ω such that the action of G on Ω is faithful, 3-transitive, the action of G_ω on S is of general type for every $\omega \in \Omega$, and for every $\Delta \in \Omega^{(2)}$, the action of G_Δ on S is quasi-parabolic.*

Denote by $\Omega^{\{2\}}$ the set of unordered pairs of distinct elements of Ω and by $\Delta^{\{2\}}$ the element of $\Omega^{\{2\}}$, corresponding to Δ . Denote by $G_{\Delta^{\{2\}}}$ the setwise stabilizer of $\Delta^{\{2\}}$ in G . Clearly, G_Δ is a subgroup of index 2 in $G_{\Delta^{\{2\}}}$, hence is normal. For the sake of contradiction, assume that the action of G_Δ on S is quasi-parabolic. Then, by the normality of G_Δ in $G_{\Delta^{\{2\}}}$, the action of $G_{\Delta^{\{2\}}}$ on S is also quasi-parabolic. Note that, by Proposition 3.16, $\Lambda_S(G) = \Lambda_S(G_\omega)$. We denote by $q(\Delta)$ the unique point of $\Lambda_S(G)$, which is fixed by G_Δ and $G_{\Delta^{\{2\}}}$. Note that G_Δ acts transitively on $\Omega \setminus \Delta$ by the 3-transitivity of the G -action on Ω , and $G_\Delta < G_{q(\Delta)}$, so $G_{q(\Delta)}$ acts transitively on $\Omega \setminus \Delta$.

The proof of Proposition 3.20 will consist of a series of lemmas that will lead to a contradiction.

Lemma 3.21. *Suppose that G and G_ω satisfy the conditions of Proposition 3.20 for any $\omega \in \Omega$. Then for every nonempty open subset $U \subset \Lambda_S(G)$, all orbits of $\text{Rist}_G(U)$ in Ω have cardinality at least 3.*

Proof. Assume for contradiction that $\text{Rist}_G(U)$ has an orbit of cardinality at most 2. Then the induced permutation group on this orbit is abelian, and the commutator subgroup $[\text{Rist}_G(U), \text{Rist}_G(U)]$ must fix a point $\omega \in \Omega$. Since the action of G_ω on S is of general type, by Lemma 2.34, for any nonempty non-dense open $V \subset \Lambda_S(G)$ there exists $f \in G_\omega$ such that $f(V) \subset U$. Thus, we have

$$f \text{Rist}_G(V) f^{-1} \leq \text{Rist}_G(U),$$

and

$$f[\text{Rist}_G(V), \text{Rist}_G(V)] f^{-1} \leq [\text{Rist}_G(U), \text{Rist}_G(U)] \leq G_\omega.$$

Since $f \in G_\omega$, we obtain $[\text{Rist}_G(V), \text{Rist}_G(V)] \leq G_\omega$.

By Lemma 2.35, the action of G on $\Lambda_S(G)$ is micro-supported, and, by Lemma 2.31, rigid stabilizers of nonempty open subsets of $\Lambda_S(G)$ are non-abelian. Therefore,

$$N = \langle [\text{Rist}_G(V), \text{Rist}_G(V)] \mid V \subset \Lambda_S(G) \text{ is a nonempty non-dense open subset} \rangle$$

is a non-trivial normal subgroup of G . Thus, G_ω contains a non-trivial normal subgroup of G . Since the action of G on Ω is 2-transitive, this means that the action of G_ω is transitive on Ω , which contradicts that G_ω fixes $\omega \in \Omega$. $\square$

Lemma 3.22. *$G_{\Delta^{\{2\}}} \neq G_{q(\Delta)}$ for any $\Delta^{\{2\}} \in \Omega^{\{2\}}$.*

Proof. Fix any $\Delta^{\{2\}} = \{\omega', \omega''\} \in \Omega^{\{2\}}$. The orbit $G_{\Delta^{\{2\}}} \omega'$ in Ω is of size at most 2. For any nonempty open subset $U \subset \Lambda_S(G)$, not containing $q(\Delta)$, the subgroup $\text{Rist}_G(U)$ fixes $q(\Delta)$, and hence $\text{Rist}_G(U) \leq G_{q(\Delta)}$. By Lemma 3.21, any orbit of $\text{Rist}_G(U)$ in Ω is of size at least 3. Therefore, $G_{\Delta^{\{2\}}} \neq G_{q(\Delta)}$. $\square$

Define $\pi: \Omega^{\{2\}} \longrightarrow \Lambda_S(G)$ by $\Delta^{\{2\}} \mapsto q(\Delta)$. This map is G -equivariant.

Lemma 3.23. *The fiber $\pi^{-1}(q(\Delta))$ forms a partition $\mathcal{P}_\Delta$ of Ω into blocks of size 2, and $G_{q(\Delta)}$ preserves $\mathcal{P}_\Delta$ and acts 2-transitively on its blocks.*

Proof. Fix $\Delta \in \Omega^{\{2\}}$. We first argue that two distinct $\Delta_1^{\{2\}}, \Delta_2^{\{2\}} \in \pi^{-1}(q(\Delta))$ are always disjoint. Assume that this is not the case. Then $\Delta_1^{\{2\}} \cap \Delta_2^{\{2\}}$ is a singleton, say $\Delta_1^{\{2\}} \cap \Delta_2^{\{2\}} = \{\omega\}$. Let $\Delta_1^{\{2\}} = \{\omega, \omega_1\}$ and $\Delta_2^{\{2\}} = \{\omega, \omega_2\}$. Since the action of G_ω on $\Omega \setminus \{\omega\}$ is 2-transitive, the actions of G_{Δ_1} and G_{Δ_2} on $\Omega \setminus \{\omega, \omega_1\}$ and $\Omega \setminus \{\omega, \omega_2\}$ are transitive, and we claim that

$$G_\omega = \langle G_{\Delta_1}, G_{\Delta_2} \rangle.$$

Indeed, it is obvious that $\langle G_{\Delta_1}, G_{\Delta_2} \rangle \leq G_\omega$, so we need to show that $G_\omega \leq \langle G_{\Delta_1}, G_{\Delta_2} \rangle$. Take an arbitrary $h \in G_\omega$. If $h\omega_i = \omega_i$, for some $i = 1, 2$, then $h \in G_{\Delta_i}$. Suppose $h\omega_i = \hat{\omega}_i \neq \omega_i$ for both $i = 1, 2$. Without loss of generality, assume $h\omega_1 = \hat{\omega} \neq \omega_1$. Then there exists $h_1 \in G_{\Delta_1}$ such that $h_1\hat{\omega} = \omega'$, where $\omega' \notin \{\omega_1, \omega_2\}$. Thus, we can find $h_2 \in G_{\Delta_2}$, such that $h_2\omega' = \omega_1$. Therefore, $h_2h_1h\omega_1 = \omega_1$, and $h \in \langle G_{\Delta_1}, G_{\Delta_2} \rangle$. Since $\Delta_1^{\{2\}}, \Delta_2^{\{2\}} \in \pi^{-1}(q(\Delta))$, subgroups G_{Δ_1} and G_{Δ_2} both fix $q(\Delta)$. Thus, it follows that the subgroup G_ω also fixes $q(\Delta)$. This contradicts the assumption that the action of G_ω on $\Lambda_S(G)$ is of general type.

It remains to show that the pairs of $\pi^{-1}(q(\Delta))$ cover Ω . Let Σ be the union of these pairs. Note that Σ defines a subset of Ω , and that this subset is $G_{q(\Delta)}$ -invariant by the equivariance of the map π . Since $G_{q(\Delta)}$ is transitive on Ω , we must have $\Sigma = \Omega$.

The subgroup $G_{q(\Delta)}$ fixes $q(\Delta)$, thus it preserves $\mathcal{P}_\Delta$. The action of $G_{q(\Delta)}$ on the blocks of $\mathcal{P}_\Delta$ is transitive because the action of $G_{q(\Delta)}$ on Ω already is. In order to show that this action of $G_{q(\Delta)}$ on the blocks of $\mathcal{P}_\Delta$ is 2-transitive, we need to show that the block stabilizers act transitively, i.e., we need to show that $G_{\Delta^{\{2\}}}$ acts transitively on $\mathcal{P}_\Delta \setminus \{\Delta\}$. Since $G_\Delta < G_{\Delta^{\{2\}}}$, it is enough to show that the action of G_Δ is transitive on $\mathcal{P}_\Delta \setminus \{\Delta\}$. Indeed, take any $\Delta', \Delta'' \in \mathcal{P}_\Delta$ that are distinct from Δ . Choose $\omega' \in \Delta'$ and $\omega'' \in \Delta''$. Since Δ', Δ'' are disjoint from Δ we have $\omega', \omega'' \in \Omega \setminus \{\Delta\}$. Since G_Δ acts transitively on $\Omega \setminus \{\Delta\}$, there exists $f \in G_\Delta$ such that $f\omega' = \omega''$. Since G_Δ preserves the blocks, it follows that we must have $f\Delta' = \Delta''$, and hence G_Δ acts transitively on $\mathcal{P}_\Delta \setminus \{\Delta\}$. $\square$

By $G_{q(\Delta)}^0$ we denote the subgroup of $G_{q(\Delta)}$ consisting of elements that fix some open neighborhood $U \subset \Lambda_S(G)$ of $q(\Delta)$ pointwise, i.e.,

$$G_{q(\Delta)}^0 = \bigcup \{ \text{Rist}_G(V) \mid V \text{ is open and } q(\Delta) \notin \bar{V} \}.$$

Observe that $G_{q(\Delta)}^0$ is a non-trivial normal subgroup of $G_{q(\Delta)}$ since the action of G on $\Lambda_S(G)$ is micro-supported by Lemma 2.35.

Lemma 3.24. *For every $\Delta^{\{2\}} \in \Omega^{\{2\}}$, we have $G_{q(\Delta)}^0 G_\Delta = G_{q(\Delta)}$.*

Proof. Recall that a group action is called *primitive* if it does not preserve any non-trivial partition. By Lemma 3.23, the action of $G_{q(\Delta)}$ on the blocks of the partition $\mathcal{P}_\Delta$ is 2-transitive, thus it is primitive. Since $G_{q(\Delta)}^0$ is a normal subgroup of $G_{q(\Delta)}$, and the action of $G_{q(\Delta)}$ on the blocks of the partition $\mathcal{P}_\Delta$ is transitive and primitive, it follows that the action of $G_{q(\Delta)}^0$ on these blocks is either trivial or transitive. Otherwise, the non-trivial partition into $G_{q(\Delta)}^0$ -orbits would be preserved by the action of $G_{q(\Delta)}$.

If the action of $G_{q(\Delta)}^0$ on the blocks of $\mathcal{P}_\Delta$ were trivial, then $G_{q(\Delta)}^0$ would consist of elements of order at most two, since all blocks have size 2, the action of $G_{q(\Delta)}^0$ on Ω is faithful, and $G_{q(\Delta)}^0$ preserves the partition $\mathcal{P}_\Delta$ as a subgroup of $G_{q(\Delta)}$. This cannot happen since $G_{q(\Delta)}^0$ contains a rigid stabilizer of some nonempty open subset $V \subset \Lambda_S(G)$, disjoint from $q(\Delta)$, and, by Lemma 2.31, $\text{Rist}_G(V)$ cannot be abelian.

Therefore, this action is transitive, and it follows that $G_{q(\Delta)}^0$ has at most two orbits in Ω , and each orbit intersects each block of $\mathcal{P}_\Delta$.

Consider the subgroup $G_{q(\Delta)}^0 G_\Delta \leq G_{q(\Delta)}$. It is indeed a subgroup since G_Δ normalizes $G_{q(\Delta)}^0$. We claim that $G_{q(\Delta)}^0 G_\Delta$ acts transitively on Ω . There are two cases to consider. First, if $G_{q(\Delta)}^0$ has only one orbit, then this is clear. Second, if $G_{q(\Delta)}^0$ has two orbits, then the subgroup G_Δ does not preserve the partition of Ω into these two orbits because G_Δ acts transitively on $\Omega \setminus \{\Delta\}$. Thus, the claim follows.

Take $\omega \in \Delta$ and any $f \in G_\omega \cap G_{q(\Delta)}$. By the equivariance of π , we have

$$q(f\Delta) = fq(\Delta) = q(\Delta),$$

thus $f\Delta$ and Δ belong to the same fiber of the partition. On the other hand, $\omega \in f\Delta \cap \Delta$ since f fixes ω . Therefore, $f\Delta = \Delta$ and

$$G_\omega \cap G_{q(\Delta)} = G_\Delta. \tag{19}$$

Recall that if a group G acts on a set Ω transitively and a subgroup $H \leq G$ also acts on Ω transitively, then for any $\omega \in \Omega$,

$$G = HG_\omega.$$

Therefore, since $G_{q(\Delta)}^0 G_\Delta$ contains a stabilizer of a point in Ω under the action of $G_{q(\Delta)}$ by (19), and both actions of $G_{q(\Delta)}$ and $G_{q(\Delta)}^0 G_\Delta$ are transitive on Ω , we finally obtain

$$G_{q(\Delta)}^0 G_\Delta = G_{q(\Delta)}. \quad \square$$

Lemma 3.25. *Fix $\Delta^{\{2\}} \in \Omega^{\{2\}}$ and $\Delta'^{\{2\}} \in \mathcal{P}_\Delta$ distinct from $\Delta^{\{2\}}$. Then there exists a nonempty open subset $V \subset \Lambda_S(G)$ such that $q(\Delta) \notin \bar{V}$ and $h \in \text{Rist}_G(V)$ such that $h\Delta^{\{2\}} = \Delta'^{\{2\}}$ and $h\Delta'^{\{2\}} \neq \Delta^{\{2\}}$.*

Proof. Consider the action of $G_{q(\Delta)}^0$ on Ω , and denote by

$$\psi: G_{q(\Delta)}^0 \longrightarrow \text{Sym}(\mathcal{P}_\Delta)$$

the induced action on the set of blocks of $\mathcal{P}_\Delta$. We have a short exact sequence

$$1 \longrightarrow \ker(\psi) \longrightarrow G_{q(\Delta)}^0 \longrightarrow \psi(G_{q(\Delta)}^0) \longrightarrow 1.$$

Note that $\ker(\psi)$ is an elementary abelian 2-group, since $G_{q(\Delta)}^0$ preserves the partition $\mathcal{P}_\Delta$, which consists of blocks of size 2.

The subgroup $G_{q(\Delta)}^0$ contains a rigid stabilizer of some nonempty open subset $V \subset \Lambda_S(G)$ with $q(\Delta) \notin \bar{V}$, thus, by Lemma 2.37, it does not consist of elements of order not greater than 4. Therefore, it is not an extension of two elementary abelian 2-groups, and $\psi(G_{q(\Delta)}^0)$ does not consist only of elements of order 2. Thus, there exists an element of $G_{q(\Delta)}^0$ having a cycle of length at least 3 under the action of $G_{q(\Delta)}^0$ on $\mathcal{P}_\Delta$, i.e., there exist an element h_0 and distinct $\Delta_1^{\{2\}}, \Delta_2^{\{2\}}, \Delta_3^{\{2\}} \in \mathcal{P}_\Delta$ such that

$$h_0 \Delta_i^{\{2\}} = \Delta_{i+1}^{\{2\}}, i = 1, 2.$$

By the definition of $G_{q(\Delta)}^0$, there exists an open subset $V_0 \subset \Lambda_S(G)$, not containing $q(\Delta)$, such that $h_0 \in \text{Rist}_G(V_0)$.

Now, since the action of $G_{q(\Delta)}$ on $\mathcal{P}_\Delta$ is 2-transitive by Lemma 3.23, we may find $f \in G_{q(\Delta)}$ such that $f \Delta_1^{\{2\}} = \Delta^{\{2\}}$ and $f \Delta_2^{\{2\}} = \Delta'^{\{2\}}$. Let $V = fV_0$. Note that $q(\Delta) \notin V$, since f fixes $q(\Delta)$. Consider the element $h = fh_0f^{-1}$. We have $h \in \text{Rist}_G(V)$, and

$$\begin{aligned} h \Delta^{\{2\}} &= fh_0f^{-1} \Delta^{\{2\}} = fh_0 \Delta_1^{\{2\}} = f \Delta_2^{\{2\}} = \Delta'^{\{2\}}; \\ h \Delta'^{\{2\}} &= fh_0f^{-1} \Delta'^{\{2\}} = fh_0 \Delta_2^{\{2\}} = f \Delta_3^{\{2\}} \neq \Delta^{\{2\}}. \end{aligned}$$

Therefore, h satisfies the conclusion. $\square$

Lemma 3.26. *Fix $\Delta^{\{2\}} \in \Omega^{\{2\}}$. Then for every nonempty open subset $U \subset \Lambda_S(G)$ there exists $g \in \text{Rist}_G(U)$ such that g exchanges the two elements of $\Delta^{\{2\}}$.*

Proof. Choose an element $f \in G$ such that f exchanges the two elements of $\Delta^{\{2\}}$. By Lemma 3.24, there exist $h \in G_{q(\Delta)}^0$ and $h' \in G_\Delta$ such that $f = hh'$. So the element h also exchanges the two elements of $\Delta^{\{2\}}$, and since $h \in G_{q(\Delta)}^0$ there exists a nonempty open subset $V \subset \Lambda_S(G)$ such that $q(\Delta) \notin \bar{V}$ and $h \in \text{Rist}_G(V)$.

Now let U be an arbitrary nonempty open subset of $\Lambda_S(G)$. Since the action of G_Δ on $\Lambda_S(G)$ is quasi-parabolic, by Lemma 2.34 there exists $l \in \mathcal{L}_S(G_\Delta)$ such that $lV \subset U$ and, thus, $g = lhl^{-1} \in \text{Rist}_G(U)$. Note that g also exchanges the two elements of $\Delta^{\{2\}}$ since $l \in G_\Delta$. $\square$

We complete the proof of Proposition 3.20.

Proof of Proposition 3.20. Let $\Delta^{\{2\}} \in \Omega^{\{2\}}$ and $\Delta'^{\{2\}} \in \mathcal{P}_\Delta$ distinct from $\Delta^{\{2\}}$, $V \subset \Lambda_S(G)$ and $h \in \text{Rist}_G(V)$ be as in Lemma 3.25, and let $\tilde{\Delta}^{\{2\}} \in \Omega^{\{2\}}$ such that $\tilde{\Delta}^{\{2\}} \cap \Delta^{\{2\}}$ and $\tilde{\Delta}^{\{2\}} \cap \Delta'^{\{2\}}$ are nonempty. Say $\Delta^{\{2\}} = \{\omega, \omega''\}$, $\Delta'^{\{2\}} = \{\omega', \tilde{\omega}\}$ and $\tilde{\Delta}^{\{2\}} = \{\tilde{\omega}, \omega\}$, where $\omega, \omega', \tilde{\omega}$, and $\omega'' \in \Omega$ are distinct. Choose a nonempty open subset $W \subset \Lambda_S(G) \setminus V$ with $q(\Delta) \notin \overline{W}$. We apply Lemma 3.26 and find $g \in \text{Rist}_G(W)$ such that $g\tilde{\omega} = \omega$ and $g\omega = \tilde{\omega}$. Note that g fixes $q(\Delta)$, and hence g must preserve the partition $\mathcal{P}_\Delta$. Since g exchanges two elements of $\Delta^{\{2\}}$ and $\Delta'^{\{2\}}$, it follows that g actually exchanges the blocks $\Delta^{\{2\}}$ and $\Delta'^{\{2\}}$. Thus, we have

$$hg\Delta^{\{2\}} = h\Delta'^{\{2\}} \neq \Delta^{\{2\}}. \quad (20)$$

Since g and h are supported in disjoint open subsets of $\Lambda_S(G)$, they commute. Therefore,

$$hg\Delta^{\{2\}} = gh\Delta^{\{2\}} = g\Delta'^{\{2\}} = \Delta^{\{2\}},$$

which contradicts (20). $\square$

The next lemma completes the proof outlined after Proposition 3.16.

Lemma 3.27. *Assume that G acts faithfully and 3-transitively on a set Ω . Then for any $\omega \in \Omega$, the action of a subgroup G_ω on S is quasi-parabolic.*

Proof. Fix any $\omega \in \Omega$. By Proposition 3.16, the action of G_ω on S is non-elementary. Therefore, it suffices to prove that it is not of general type. For the sake of contradiction, assume that the action of G_ω on S is of general type. The action of G on Ω is 3-transitive. Therefore, the action of G_ω on $\Omega \setminus \{\omega\}$ is 2-transitive. Let $\omega_1 \in \Omega \setminus \{\omega\}$ and $\Delta = (\omega_1, \omega)$. We apply Proposition 3.16 to $G_\Delta \leq G_\omega$ and obtain that G_Δ is itself non-elementary. By Proposition 3.19, the action of G_Δ on S cannot be of general type, and by Proposition 3.20, it cannot be quasi-parabolic. This leads to a contradiction. $\square$

Now we are ready to prove Theorem I.

Proof of Theorem I. By Lemma 3.27, the action of G_ω on S is quasi-parabolic for any $\omega \in \Omega$. Let $q(\omega) \in \Lambda_S(G_\omega)$ be the unique fixed point of G_ω . Consider the map $q: \Omega \rightarrow \partial S$, given by $\omega \mapsto q(\omega)$. This map is G -equivariant. For the sake of contradiction, assume that q is not injective, i.e., there exists $\omega_1, \omega_2 \in \Omega$, such that $\omega_1 \neq \omega_2$, but $q(\omega_1) = q(\omega_2)$. The action of G is 2-transitive on Ω , so for any distinct $\omega'_1, \omega'_2 \in \Omega$ there exists $g \in G$ such that $g\omega_1 = \omega'_1$ and $g\omega_2 = \omega'_2$. Therefore, $G_{\omega_1} = g^{-1}G_{\omega'_1}g$ and $G_{\omega_2} = g^{-1}G_{\omega'_2}g$. Thus, $q(\omega'_1) = gq(\omega_1) = gq(\omega_2) = q(\omega'_2)$. Then the map q is constant, and it follows that G fixes a point in $\Lambda_S(G)$, which contradicts the assumption that the action of G on S is of general type. Therefore, q is injective, and the G -action on Ω is conjugate to the action on $\mathcal{O} = q(\Omega)$. $\square$

Proof of Corollary J. The group G admits a general type action on a hyperbolic space S such that the induced action of G on $\Lambda_S(G)$ is faithful and not topologically free. Thus, by Theorem I, if G acts faithfully and 3-transitively on any set Ω , then this action is conjugate to the action of G on some G -orbit $\mathcal{O} \subset \Lambda_S(G)$. By Lemma 2.9, the set of fixed points of elements of $\mathcal{L}_S(G)$ is dense in $\Lambda_S(G)$. Thus, since $\mathcal{O}$ is G -invariant, by Lemma 2.6, it is also dense in $\Lambda_S(G)$. Therefore, by Proposition 3.7, the action of G on $\mathcal{O}$ is not topologically 4-transitive. Thus, by Lemma 3.9, it is at most 3-transitive. Therefore, G does not admit any faithful 4-transitive action on any set Ω , and the transitivity degree of G is at most 3. $\square$

Proof of Corollary K. Since G is faithful weakly hyperbolic and not lim-free, it admits a general type action on a hyperbolic space S such that the induced action of G on $\Lambda_S(G)$ is faithful and not topologically free. Moreover, by Theorem B, for any general type action of G on any hyperbolic space, such that the induced action on the limit set is faithful, the induced action on the limit set is not topologically free. Since the transitivity degree of G is 3, there exists a set Ω such that the G -action on Ω is faithful and 3-transitive. Then, by Theorem I, the G -action on Ω is conjugate to the G -action on some orbit $\mathcal{O} \subset \Lambda_S(G)$. Since the action of G on S is of general type, $\mathcal{O}$ is dense in $\Lambda_S(G)$. Therefore, the action of G on $\Lambda_S(G)$ is topologically 3-transitive. $\square$

Corollary 3.28. [BM, Theorem 1.4] *Let T be a simplicial tree. Suppose that the action of $G \leq \text{Aut}(T)$ on T is minimal and of general type and that the action of G on ∂T is not topologically free. Assume that G acts faithfully and 3-transitively on a set Ω . Then there exists a G -orbit $\mathcal{O} \subset \partial T$ such that the actions of G on Ω and on $\mathcal{O}$ are conjugate.*

Proof. It is sufficient to show that G satisfies the conditions of Theorem I. This was already shown in the proof of Corollary 3.14. $\square$

3.4 Examples and necessity of assumptions

Example 3.29. Verbal products. This example provides a family of non-faithful weakly hyperbolic groups that satisfy a mixed identity.

Let $V = \{f_\nu(x_1 \dots x_{n_\nu}) : \nu \in I\}$ be a subset of words in an infinitely generated free group. A verbal V -subgroup $V(G)$ of a group G is a subgroup generated by all possible values of all words of V , when $x_1, x_2, \dots$ run through the entire group G independently of each other. A verbal V -subgroup $V(G)$ is normal in G . Let $G_i, i \in I$ be some collection of groups and let $F = \ast_{i \in I} G_i$. Let $\pi: F \rightarrow \prod_{i \in I} G_i$ be the natural epimorphism. Then the Cartesian subgroup $C < F$ is the kernel of π . Let $V(F)$ be a verbal V -subgroup of F . Then the verbal product $\ast_V G_i$ of groups G_i with respect to V is $F/(V(F) \cap C)$.

Lemma 3.30. *Let A and B be nontrivial groups, and let V be a nonempty subset of words in a free group $F(x_1, \dots, x_k)$. Then the group $G = A \ast_V B$ satisfies a mixed identity.*

Proof. Let $v = v(x_1, x_2, x_3, \dots, x_k)$ be a nonempty word in V , and let $a \in A$ and $b \in B$ be non-trivial elements of G . Consider the element

$$w(a, b, x_1, x_2, x_3, \dots, x_k) = v([a^{x_1}, b], [a^{x_2}, b], [a^{x_3}, b], \dots, [a^{x_k}, b]) \in G * F(x_1, \dots, x_k).$$

For any $i \in \{1, \dots, k\}$ and values of x_i in G , elements $[a^{x_i}, b]$ are in the Cartesian subgroup of $A * B$, thus $v([a^{x_1}, b], [a^{x_2}, b], [a^{x_3}, b], \dots, [a^{x_k}, b]) \in V(A * B)$. Therefore, $w(a, b, g_1, g_2, g_3, \dots, g_k) = 1$ in G for any values $g_1, \dots, g_k \in G$. Since a, b are non-trivial in G , $w(a, b, x_1, x_2, x_3, \dots, x_k)$ is non-trivial in $G * F(x_1, \dots, x_k)$. $\square$

Proposition 3.31. *Let A, B be nontrivial groups, and let at least one of A or B be weakly hyperbolic. Let V be a nonempty subset of words in a free group $F(x_1, \dots, x_k)$. Then $G = A *_V B$ is a weakly hyperbolic group that is not faithful weakly hyperbolic.*

Proof. Without loss of generality, assume that A is weakly hyperbolic. There exists a surjective homomorphism of G onto A , thus G is also weakly hyperbolic. Let $\varphi(C)$ be the image of the Cartesian subgroup $C < A * B$ under the homomorphism $\varphi: A * B \rightarrow G$.

If $\varphi(C) = 1$, then G is a direct product of two non-trivial groups, so it is not MIF and $\text{Mon}(G) = 1$. Thus, by Proposition D, G cannot be faithful weakly hyperbolic.

Assume now that $\varphi(C) \neq 1$. Since φ is surjective, $\varphi(C)$ is normal in G . Therefore, by Lemma 2.10, for any general type action of G on any hyperbolic space S , the subgroup $\varphi(C)$ is either elliptic or of general type. Since $V(\varphi(C)) = \{1\}$, $\varphi(C)$ satisfies some non-trivial identity $v = 1$, $v \in V$ and cannot contain a non-abelian free subgroup. So, $\varphi(C)$ is elliptic, and since it is normal in G , $\varphi(C) < E_S(G)$ by the maximality of $E_S(G)$. Therefore, $E_S(G)$ is non-trivial, and G is not a faithful weakly hyperbolic group. $\square$

Example 3.32. Non-lim-free HNN extensions and amalgams. The following proposition provides restrictions on associate subgroups of HNN-extensions and amalgamated products when the action on the boundary of the Bass-Serre tree is not topologically free.

Definition 3.33. Let H be a group and let $\varphi: C \rightarrow \varphi(C)$ be an isomorphism between subgroups of H . Then HNN-extension associated to (H, C, φ) is the group defined by the following presentation

$$\text{HNN}(H, C, \varphi) := \langle H, t \mid t^{-1}ct = \varphi(c) \text{ for all } c \in C \rangle,$$

where t is an extra generator, not belonging to H , called a *stable letter*.

Recall that it is called *ascending* if one of the subgroups $C, \varphi(C)$ is equal to H .

Definition 3.34. Let $l_1: C \rightarrow A$ and $l_2: C \rightarrow B$ be injective group homomorphisms. We will denote by C_j the image of l_j and by $\varphi: C_1 \rightarrow C_2$ the isomorphism sending $l_1(c)$ to $l_2(c)$ for all $c \in C$. The *free product with amalgamation (or amalgam for short)* associated to (A, B, C, φ) is the group defined by the following presentation:

$$A *_C B := \langle A, B \mid l_1(c) = l_2(c) \text{ for all } c \in C \rangle = \langle A, B \mid c = \varphi(c) \text{ for all } c \in C_1 \rangle.$$

Recall that such an amalgam is said to be *non-trivial* if $A, B \neq C_j$ for $j = 1, 2$, and *non-degenerate* if moreover $[A : C_1] \geq 3$ or $[B : C_2] \geq 3$.

Given an HNN-extension $G = \text{HNN}(H, C, \varphi)$ or an amalgam $G = A *_C B$, we can construct a Bass-Serre tree T associated to G . The group G acts on T transitively on the edges of T . Stabilizers of edges are conjugates of C or $\varphi(C)$. The action of G on T is always minimal. We refer the reader to [Ser] for more information. If $\text{HNN}(H, C, \varphi)$ is non-ascending or $G = A *_C B$ is non-degenerate, then the action of G on T is of general type (see, for example, [FMMS, Sections 2.5, 2.6]).

Proposition 3.35. *Let $G = \text{HNN}(H, C, \varphi)$ be a non-ascending HNN-extension or a non-degenerate amalgam $G = A *_C B$. If the induced action of G on the boundary ∂T of the Bass-Serre tree is faithful but not topologically free, then C is non-abelian.*

Proof. By Lemma 2.35, the action of G on ∂T is micro-supported, and, by Lemma 2.31, $\text{Rist}_G(U)$ is non-abelian for every nonempty open subset $U \subseteq \partial T$. Take an arbitrary open U that is not dense in ∂T . Consider two adjacent vertices v_1 and v_2 in T , such that the ends of infinite rays from these vertices are in $\partial T \setminus U$. Then $\text{Rist}_G(U)$ fixes both v_1 and v_2 ; thus, stabilizes the edge e incident to v_1 and v_2 . Since stabilizers of edges are conjugate to C , there exists $g \in G$ such that $\text{Rist}_G(U) \leq C^g$. Therefore, C is non-abelian. $\square$

Corollary 3.36. *The action of a non-solvable Baumslag-Solitar group $BS(m, n)$ on the boundary of its Bass-Serre tree is not topologically free if and only if $|m| = |n|$.*

Example 3.37. Solvable Baumslag-Solitar groups. This example shows that the conclusion of Theorem 3.10 does not hold if we take a quasi-parabolic action instead of a general-type action in the assumptions.

The Baumslag-Solitar group $BS(1, 2)$ admits a quasi-parabolic action on its Bass-Serre tree T_{BS} since it is an ascending HNN extension (see [FMMS, Section 2.5]), such that the induced action on $\Lambda_{T_{BS}}(BS(1, 2)) = \partial T_{BS}$ is topologically free [FMMS, Lemma 8.6]. However, $BS(1, 2)$ is solvable of order 3, so it satisfies the identity

$$[[[x_1, x_2], [x_3, x_4]], [x_5, x_6], [x_7, x_8]] = 1.$$

Example 3.38. Topologically free action on the boundary. This example shows why in Theorem 3.10 we must consider topologically free actions on the limit set, not the whole boundary as an equivalent condition. A similar example is given in [FMMS, Proposition 9.2], where they discuss the necessity of minimality of the action on the tree in their case, which exactly gives faithfulness of the action on the limit set.

Let $G = F_2 \times \mathbb{Z}$, where F_2 is a free group of rank 2. The action of F_2 on its Cayley graph T is of general type. Let E be a countable bundle of rays from a unique point o , enumerated by integers. The group $\mathbb{Z}$ acts on E as follows: $n \cdot r_m = r_{n+m}$ for every $n \in \mathbb{Z}$ and every ray $r_m \subset E$. This action of $\mathbb{Z}$ on E is elliptic since it fixes a point o . Let S be a space, constructed by gluing over each vertex of T a copy of E at its fixed point o . Then G acts on

S by isometries, where elements from F_2 act on a copy of T and elements of $\mathbb{Z}$ act on copies of E in S . This action is of general type, and the induced action on ∂S is topologically free. However, it is not topologically free (and not even faithful) on $\Lambda_S(G) = \partial T$, since elements $(1, n) \in F_2 \times \mathbb{Z}$, $n \neq 0$ fix ∂T . Obviously, G satisfies a nontrivial mixed identity $[f^x, n] = 1$, where $f \in F_2$, x is a free variable, and $n \in \mathbb{Z} \setminus \{0\}$.

Example 3.39. Monolithic lim-free groups. This example shows that the converse of Proposition D is not true, i.e., there exist lim-free weakly hyperbolic (hence MIF) groups with infinite simple monolith.

Consider $\mathrm{PSL}_2(\mathbb{C})$, $\mathrm{PSL}_2(\mathbb{R})$, $\mathrm{PSL}_2(\mathbb{Q})$, or any $\mathrm{PSL}_2(F)$, where F is a field of $\mathrm{char} = 0$ with cardinality less or equal to continuum. By the proof of Corollary F, they are lim-free, and it is well known that they are simple (see [Dix]), so coincide with their monolith.

Example 3.40. Non-faithful weakly hyperbolic MIF groups.

Proposition 3.41. *Every countable group can be embedded into a finitely generated MIF weakly hyperbolic group, which is not faithful.*

Proof. Let K be an arbitrary countable group. By the Higman-Neumann-Neumann Embedding Theorem [HNN], K can be embedded in a finitely generated group G . By the construction in [HNN], G is a non-ascending HNN-extension; therefore, it is weakly hyperbolic. Then by [HL, Thm 1.1], there exists a finitely generated group H , such that:

1. $G \leq H \leq \mathrm{Sym}(G \times \mathbb{Z})$;
2. There is a short exact sequence:

$$1 \longrightarrow N \longrightarrow H \longrightarrow (G \times \mathbb{Z}) \longrightarrow 1,$$

where N is locally finite;

3. H is MIF.

The group H maps surjectively on $G \times \mathbb{Z}$, and $G \times \mathbb{Z}$ is weakly hyperbolic, since G is weakly hyperbolic. Therefore, H also admits a general type action on the same hyperbolic space as G . Note that the subgroup N is non-trivial; otherwise H would be isomorphic to $G \times \mathbb{Z}$, which is impossible since H is MIF and $G \times \mathbb{Z}$ has a non-trivial center. Since N is locally finite, it consists of torsion elements of H . Since N is normal in H , it cannot be a parabolic subgroup with respect to any action on any hyperbolic space, so it is always elliptic, and, by Proposition 2.24, it should be in an elliptic radical of H with respect to any general type action. Therefore, any elliptic radical of H is non-trivial, and H is not faithful weakly hyperbolic. $\square$

Example 3.42. Finitely presented simple MIF groups.

Kac-Moody groups. We partially answer Question 3.6 from [BFFHZ].

Proof of Proposition H. Let $\Gamma(F)$ be a finitely presented Kac-Moody group over a finite field, constructed in [CR], such that its quotient by the finite center $G(F) = \Gamma(F)/Z(\Gamma(F))$ is an infinite finitely presented simple group. We want to show that $G(F)$ is lim-free weakly hyperbolic. First, we will show that $G(F)$ is weakly hyperbolic.

The group $G(F)$ is a nonuniform lattice in a product of two buildings. Since F is finite, the associated building is locally finite; hence its Davis realization X is a proper CAT(0) space. Since F is a field, by the proof of [CF, Corollary 1.3 (page 19, line 1 of the proof)], $G(F)$ acts Weyl-transitively on each of the two buildings. By [CF, Proposition 5.3], the action of $G(F)$ on one of these buildings X contains two rank-one elements g_1 and g_2 with disjoint limit points in the boundary of X .

By [PSZ, Theorem B], there exists δ such the curtain model X_D associated to X is δ -hyperbolic and $\text{Isom}(X) \leq \text{Isom}(X_D)$. By [PSZ, Theorem C], two rank-one elements g_1 and g_2 of the action of $G(F)$ on X become loxodromic elements of the action of $G(F)$ on X_D . Finally, by [PSZ, Theorem L], the boundary ∂X_D embeds as an Isom-invariant subset of ∂X ; thus, the four limit points of rank-one elements g_1 and g_2 in ∂X must also be four distinct points when looking at the corresponding limit points on ∂X_D . Therefore, the action of $G(F)$ on X_D is of general type. Since $G(F)$ is simple, $E_{X_D}(G(F)) = \{1\}$, and the action of $G(F)$ on $\Lambda_{X_D}(G(F))$ is faithful. Note that the space X_D is not necessarily geodesic, though, by [PSZ, Proposition N], there is a geodesic hyperbolic space $E(X_D)$, called the *injective hull*, such that the canonical embedding $X_D \rightarrow E(X_D)$ is an $\text{Isom}(X_D)$ -equivariant quasi-isometry. In particular, it induces an $\text{Isom}(X)$ -equivariant homeomorphism between the Gromov boundaries of X_D and $E(X_D)$. Thus, g_1 and g_2 act loxodromically on $E(X_D)$ and have disjoint pairs of fixed points. Moreover, the elliptic radical

$$E_{E(X_D)}(G(F)) = E_{X_D}(G(F)) = \{1\}.$$

Thus, we can pass to the space $E(X_D)$ to conclude that $G(F)$ is faithful weakly hyperbolic.

Now, assume that the action of $G(F)$ on $\Lambda_{E(X_D)}(G(F))$ is not topologically free. Then, by Corollary 2.36, the rigid stabilizers of nonempty open subsets of $\Lambda_{E(X_D)}(G(F))$ are infinite and contain infinite direct sums of other rigid stabilizers. Assume that a rigid stabilizer of some nonempty non-dense open subset U of $\Lambda_{E(X_D)}(G(F))$ has an element of order $1 < n < \infty$. Then Lemma 2.37 allows us to construct elements of arbitrarily large finite order in $\text{Rist}_{G(F)}(U)$. Since the center $Z(\Gamma(F))$ is finite, this contradicts [Cap, Corollary 1.3]. Therefore, the rigid stabilizers of nonempty non-dense open subsets in the limit set $\Lambda_{E(X_D)}(G(F))$ are torsion-free. Thus, $G(F)$ contains an infinite direct sum P of infinite cyclic subgroups. By [BD, Theorem 64], the asymptotic dimension of $G(F)$ is at least the supremum of the asymptotic dimensions of its finitely generated subgroups. For every $n \in \mathbb{N}$, a subgroup P contains a direct sum P_n isomorphic to $\mathbb{Z}^n$. Since the asymptotic dimension of $\mathbb{Z}^n$ is n , we obtain that the asymptotic dimension of $G(F)$ is infinite. This yields a contradiction with the fact that the asymptotic dimension of $G(F)$ is bounded from above by the asymptotic dimension of the product of two buildings it acts on [SWZ, last paragraph], which is finite by [DS]. Therefore, the action of $G(F)$ on $\Lambda_{E(X_D)}(G(F))$ is

topologically free and, by Theorem B, $G(F)$ is MIF. By Corollary G, the reduced C^* -algebra of $G(F)$ is selfless. $\square$

Burger-Mozes group. In [BM97], the finitely presented simple Burger-Mozes group G was constructed. It acts by isometries on a locally finite tree T such that the induced action on ∂T is 2-transitive ([BM97, Remark 5]). Therefore, the action of G on T is of general type. By construction, G acts properly and cocompactly by isometries on the product of two locally finite trees. Hence, it has finite asymptotic dimension. So, the argument of Proposition H can be applied to G to conclude that G is MIF, and the reduced C^* -algebra of G is selfless. The fact that G is MIF was first observed in [BFFHZ, Remark 3.5].

Amir-Lazarovich groups. Very recently, in [AL], M. Amir and N. Lazarovich constructed simple finitely presented groups acting on products of two Davis complexes: a c -regular tree with $c \geq 6$ and the right-angled Davis complex of an Odd graph. By [AL, Theorem 5.2], the action of such groups on a tree is vertex-transitive and therefore is of general type. Thus, the rest of the argument of Proposition H also applies, and such groups are MIF and have selfless reduced C^* -algebras.

Remark 3.43. In [TMW] a family of lattices acting on two-dimensional Euclidean buildings was introduced, and recently it was proved that this construction also produces a family of infinite simple groups [LW]. Unfortunately, the reasoning from Proposition H does not apply to this family, and we do not know whether they are mixed identity free or not.

References

- [ABDHMW] T. Aougab, H. Bray, S. Dowdall, H. Hoganson, S. Maloni, B. Whitfield, Constructing reducibly geometrically finite subgroups of the mapping class group. *Groups Geom. Dyn.*, 2026.
- [AG] N. Avni, T. Gelander, Mixed identities in linear groups – effective version, *arXiv:2510.03492v1*.
- [AGKEP] T. Amrutam, D. Gao, S. Kunawalkam Elayavalli, G. Patchell, Strict comparison in reduced group C^* -algebras. *Invent. math.* **242** (2025), 639–657.
- [AL] M. Amir, N. Lazarovich, Simple lattices in products of Davis complexes, *arXiv:2605.09493*.
- [AMS] R. Aoun, P. Mathieu and Ç. Sert, Random walks on hyperbolic spaces: second order expansion of the rate function at the drift, *J. Éc. polytech. Math.* **10** (2023), 549–573.
- [AS] R. Aoun and Ç. Sert, Random walks on hyperbolic spaces: concentration inequalities and probabilistic Tits alternative, *Probab. Theory Related Fields* **184** (2022), no. 1-2, 323–365.

- [Bas] H. Bass, Covering theory for graphs of groups, *J. Pure Appl. Algebra* **89** (1993), no. 1-2, 3–47.
- [BD] G. Bell, A. Dranishnikov, Asymptotic dimension, *Topology Appl.*, **155** (2008), 1265–1296.
- [Bea] A. Beardon, The geometry of discrete groups. Graduate Texts in Mathematics, *Springer*, **91**, New York etc., 1995.
- [BEH] C. Bleak, L. Elliott, J. T. Hyde, Sufficient conditions for a group of homeomorphisms of the Cantor set to be two-generated, *J. Inst. Math. Jussieu* **23** (2024), no. 6, 2825–2858.
- [Bes] M. Bestvina. $\mathbb{R}$ -trees in topology, geometry, and group theory. In *Handbook of Geometric Topology*, pages 55-91. North Holland, 2001.
- [BFFHZ] J. Belk, F. Fournier-Facio, J. Hyde, M. C. B. Zaremsky, Boone-Higman embeddings of $\text{Aut}(F_n)$ and mapping class groups of punctured surfaces, *arXiv:2503.21882v2*.
- [BFGS] S. Balasubramanya, F. Fournier-Facio, A. Genevois, Property (NL) for hyperbolic group actions, with Appendix by A. Sisto, *arXiv:2212.14292*.
- [BH] H. Baik, W. Jang, On the kernel of actions on asymptotic cones, *Journal of Group Theory*, **28** (2025) 753–809.
- [BH99] M.R. Bridson, A. Haefliger, Metric spaces of non-positive curvature. *Grundlehren der Mathematischen Wissenschaften* [Fundamental Principles of Mathematical Sciences] **319**. Springer-Verlag, Berlin, 1999.
- [BM] A. Le Boudec, N. Matte Bon, Triple transitivity and non-free actions in dimension one, *J. Lond. Math. Soc.* **105** (2022), 2, 884–908.
- [BM97] M. Burger and S. Mozes, Finitely presented simple groups and products of trees, *C. R. Acad. Sci. Paris Sér. I Math.*, (1997) **324** 747–752.
- [BMSS] A. Boulanger, P. Mathieu, C. Sert, A. Sisto, Large deviations for random walks on Gromov-hyperbolic spaces, *Ann. Sci. Éc. Norm. Supér. (4)* **56** (2023), no. 3, 885–944.
- [BS07] Sergei Buyalo and Viktor Schroeder. Elements of asymptotic geometry. *EMS Monographs in Mathematics. European Mathematical Society (EMS)*, Zurich, 2007.
- [BS] H. Bradford, A. Sisto, Non-solutions to mixed equations in acylindrically hyperbolic groups coming from random walks, *arXiv:2504.15456v1*.
- [CDP] M. Coornaert, T. Delzant, A. Papadopoulos, Géométrie et théorie des groupes, *Lecture Notes in Mathematics*, **1441**, Springer-Verlag, Berlin, 1990.

- [Cap] P.-E. Caprace, A uniform bound on the nilpotency degree of certain subalgebras of Kac–Moody algebras, *J. of Algebra* **317**(2) (2007), 867–876.
- [CF] P.-E. Caprace, K. Fujiwara, Rank-One Isometries of Buildings and Quasi-Morphisms of Kac–Moody Groups. *Geom. Funct. Anal.* **19** (2010), 1296–1319.
- [Cho] I. Choi, Central limit theorem and geodesic tracking on hyperbolic spaces and Teichmüller spaces, *Adv. Math.*, **431** (2023), 68.
- [Cou16] R. Coulon, Partial periodic quotients of groups acting on a hyperbolic space, *Ann. Inst. Fourier (Grenoble)*, **66** (2016), 1773–1857.
- [CR] P.-E. Caprace, B. Rémy, Simplicity and superrigidity of twin building lattices, *Invent. Math.*, **176**(2009), 169–221.
- [CRW] P. Caprace, C. Reid, G. Willis, Locally normal subgroups of totally disconnected groups. Part II: Compactly generated simple groups, *Forum Math. Sigma* **5** (2017), no. e12, 89.
- [DGO] F. Dahmani, V. Guirardel, and D. Osin, Hyperbolically embedded subgroups and rotating families in groups acting on hyperbolic spaces, *Mem. Amer. Math. Soc.*, **245** (2017), v+152.
- [Dix] L. E. Dickson, Theory of linear groups in an arbitrary field, *Trans. Amer. Math. Soc.* **2** (1901), 363–394.
- [DSU] T. Das, D. Simmons, M. Urbanski, Geometry and dynamics in Gromov hyperbolic metric spaces: With an emphasis on non-proper settings, *Mathematical Surveys and Monographs* **218**, 2017.
- [FMMS] P. Fima, F. Le Maître, S. Moon, Y. Stalder, A characterization of high transitivity for groups acting on trees, *Discrete Anal.* **8** (2022), 63.
- [GGS] T. Gelander, Y. Glasner, G. Soifer, Maximal subgroups of countable groups: a survey. In *Dynamics, geometry, number theory: the impact of Margulis on modern mathematics*, *University of Chicago Press* (2022), 169–210.
- [GOR] G. Goffar, D. V. Osin and E. Rybak, Frattini subgroups of hyperbolic-like groups, *J. Algebra* **658** (2024), 22–37.
- [Gro87] M. Gromov, Hyperbolic groups, *Essays in Group Theory*, *MSRI Series* **8**, (S.M. Gersten, ed.), *Springer*, 1987, 75–263.
- [GR] A. Garrido, C. D. Reid, Compressible subgroups and simplicity, *arxiv:2412.18891*
- [DS] J. Dymara, T. Schick, Buildings have finite asymptotic dimension, *Russ. J. Math. Phys.*, **16** (2009), 409–412.

- [GTT22] I. Gekhtman, S. J. Taylor and G. Tiozzo, Central limit theorems for counting measures in coarse negative curvature, *Compos. Math.* **158** (2022), no. 10, 1980–2013.
- [Ham] M. Hamann, Group actions on metric spaces: fixed points and free subgroups, *Abh. Math. Semin. Univ. Hambg.* **87** (2017), 245–263.
- [HL] J. Hyde, Y. Lodha, Embeddings into highly transitive and mixed identity free groups, *arXiv:2509.09788*, 2025
- [HNN] G. Higman, B. H. Neumann and H. Neumann, Embedding theorems for groups, *J. London Math. Soc.* **24** (1949), 247–254.
- [HO13] M. Hull, D. Osin, Conjugacy growth of finitely generated groups, *Advances in Mathematics.* **235** (2013) (1), 361–389.
- [HO16] M. Hull, D. Osin, Transitivity degrees of countable groups and acylindrical hyperbolicity, *Israel J. Math.* **216** (2016), no.1, 307–353.
- [Hor] C. Horbez, Central limit theorems for mapping class groups and $\text{Out}(F_N)$, *Geom. Topol.* **22** (2018), no. 1, 105–156.
- [HV] P. de la Harpe, Alain Valette. La propriété (T) de Kazhdan pour les groupes localement compacts (avec un appendice de Marc Burger). *Astérisque*, No. 175. Société Mathématique de France, Paris, 1989. With an appendix by M. Burger.
- [Jac] B. Jacobson, Algebraic subgroups of acylindrically hyperbolic groups, *J. of Algebra*, **476** (2017), 113–133.
- [KM] O. Kharlampovich, A. Myasnikov, Definable sets in a hyperbolic group, *Internat. J. of Algebra and Comput.* **23** (2013), 91–110.
- [KS] M. Kapovich, P. Sardar, Trees of hyperbolic spaces, *Mathematical surveys and monographs*, **282**, 2024.
- [La] S. Lang, Algebra. Volume 1. Revised third edition, *Graduate Texts in Mathematics* **211**, New York, NY: Springer (2002)
- [LS] R. Lyndon, P. Schupp, Combinatorial Group Theory, *Classics in Mathematics*, Springer Berlin, Heidelberg (2001)
- [LW] J. Lécureux, S. Witzel, The Normal Subgroup Theorem for lattices on two-dimensional Euclidean buildings, *arXiv:2605.06163*
- [Mah] J. Maher, Linear progress in the complex of curves, *Trans. Amer. Math. Soc.* **362** (2010), 2963–2991.
- [Mal] A.I. Malcev, On isomorphic matrix representations of infinite groups of matrices (Russian), *Mat. Sb.* **8** (1940), 405–422, *Amer. Math. Soc. Transl.* (2) **45** (1965), 1–18.

- [MT] J. Maher, G. Tiozzo, Random walks on weakly hyperbolic groups, *J. Reine Angew. Math.*, **742** (2018), 187–239.
- [Osi16] D. V. Osin, Acylindrically hyperbolic groups, *Trans. Amer. Math. Soc.*, **368** (2016), pp. 851–888.
- [Osi17] D. Osin, Invariant random subgroups of groups acting on hyperbolic spaces. *Proc. Amer. Math. Soc.*, **145** (2017), no. 8, 3279–88.
- [Oza] N. Ozawa, Proximity and selflessness for group C^* -algebras, *arXiv:2508.07938v5*.
- [Pau96] F. Paulin, Un groupe hyperbolique est déterminé par son bord. *J. London Math. Soc.* **54**(1) (1996), 50–74.
- [PSZ] H. Petyt, D. Spriano, A. Zalloum, Hyperbolic models for CAT(0) spaces, *Adv. in Math.*, **450** (2024).
- [Rob] L. Robert, Selfless C^* -algebras, *Adv. Math.* **478** (2025), 110409.
- [Ser] P. Serre (with H. Bass), Trees, *Springer-Verlag*, 1980.
- [Sis] A. Sisto, Quasi-convexity of hyperbolically embedded subgroups, *Mathematische Zeitschrift*. **283** (2016) (3–4), 649–658.
- [Sun] M. Sunderland, Linear progress with exponential decay in weakly hyperbolic groups, *Groups Geom. Dyn.*, **14**(2) (2020), 539–566.
- [TMW] T. Titz Mite, S. Witzel, Non-residually finite $\tilde{C}_2$ -lattices, *arXiv:2509.05054*.
- [SWZ] R. Skipper, S. Witzel, M.C.B. Zaremsky, Simple groups separated by finiteness properties, *Invent. math.* **215** (2019), 713–740.
- [Tom] G. Tomanov, Generalized group identities in linear groups, (Russian) *Mat. Sb. (N.S.)* **1**, (1984) 35–49.
- [Vid] I. Vigdorovich, Structural properties of reduced C^* -algebras associated with higher-rank lattices, *arXiv:2503.12737v3*.

Ekaterina Rybak: Department of Mathematics, Vanderbilt University, Nashville, TN, U.S.A.

E-mail: *ekaterina.rybak@vanderbilt.edu*